

BFS Traversal

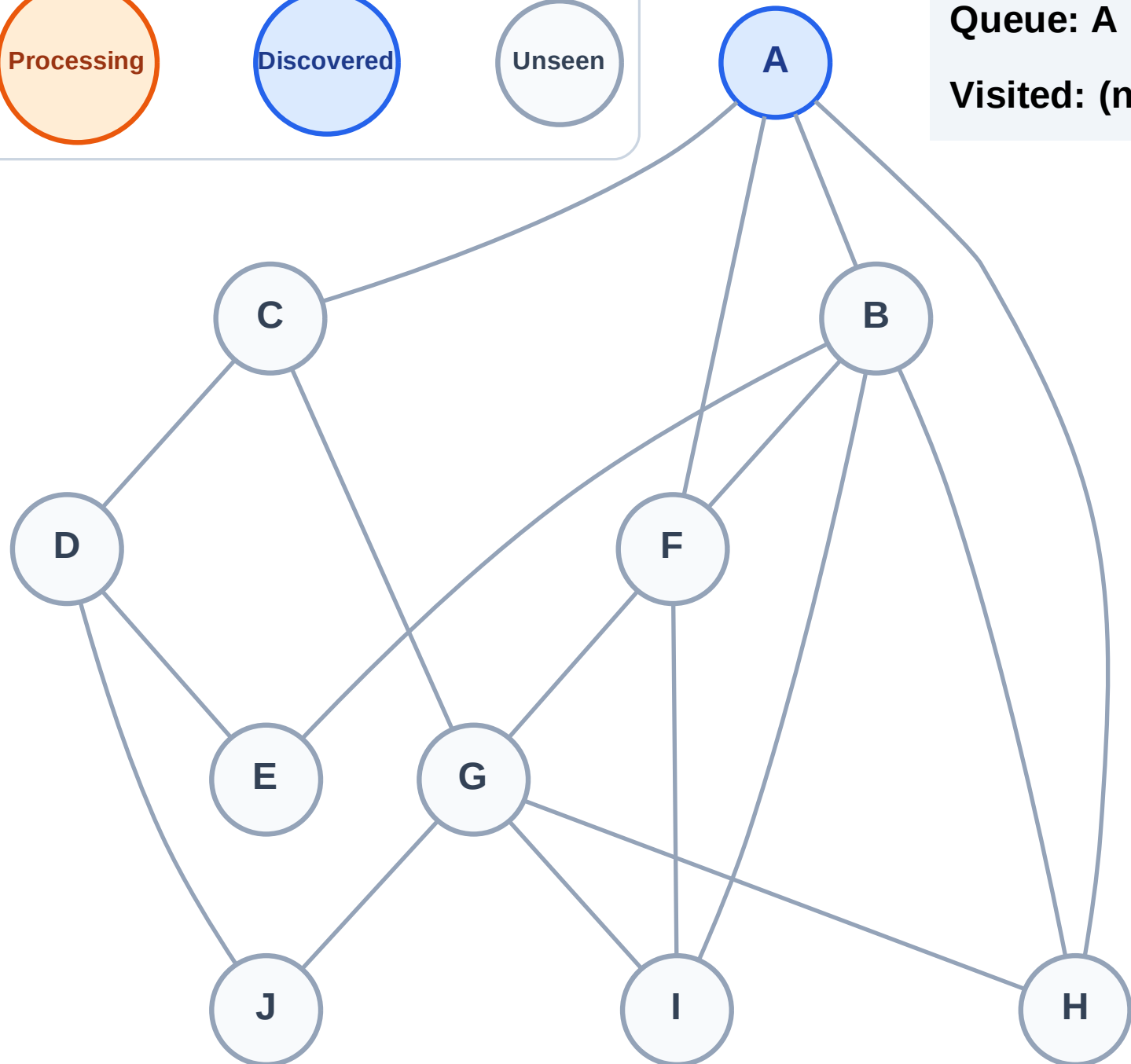
Step 1: Start at A

Legend



Queue: A

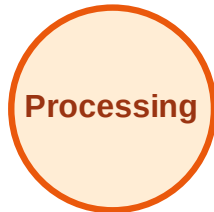
Visited: (none)



BFS Traversal

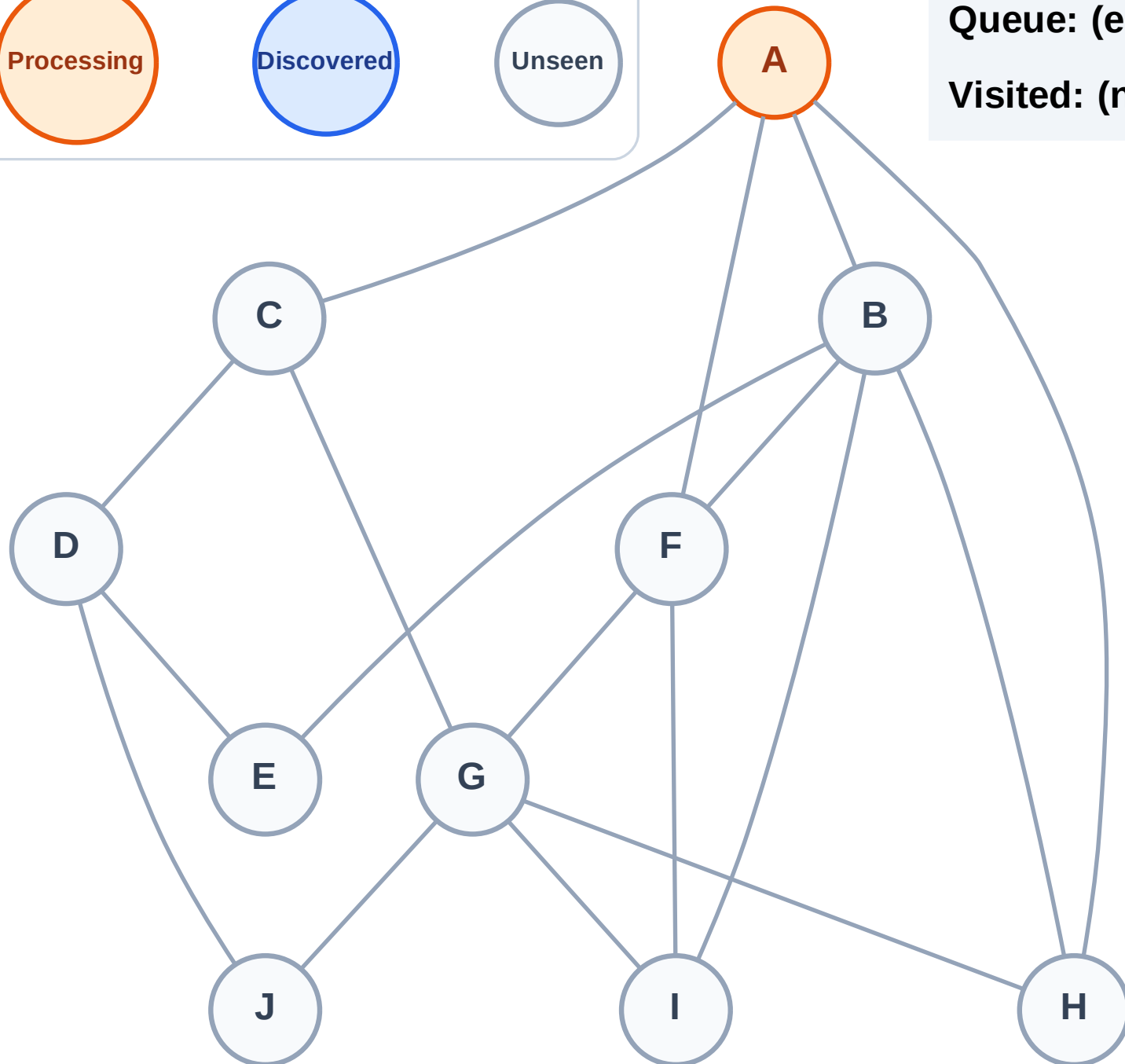
Step 2: Process node A

Legend



Queue: (empty)

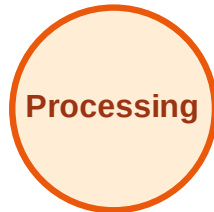
Visited: (none)



BFS Traversal

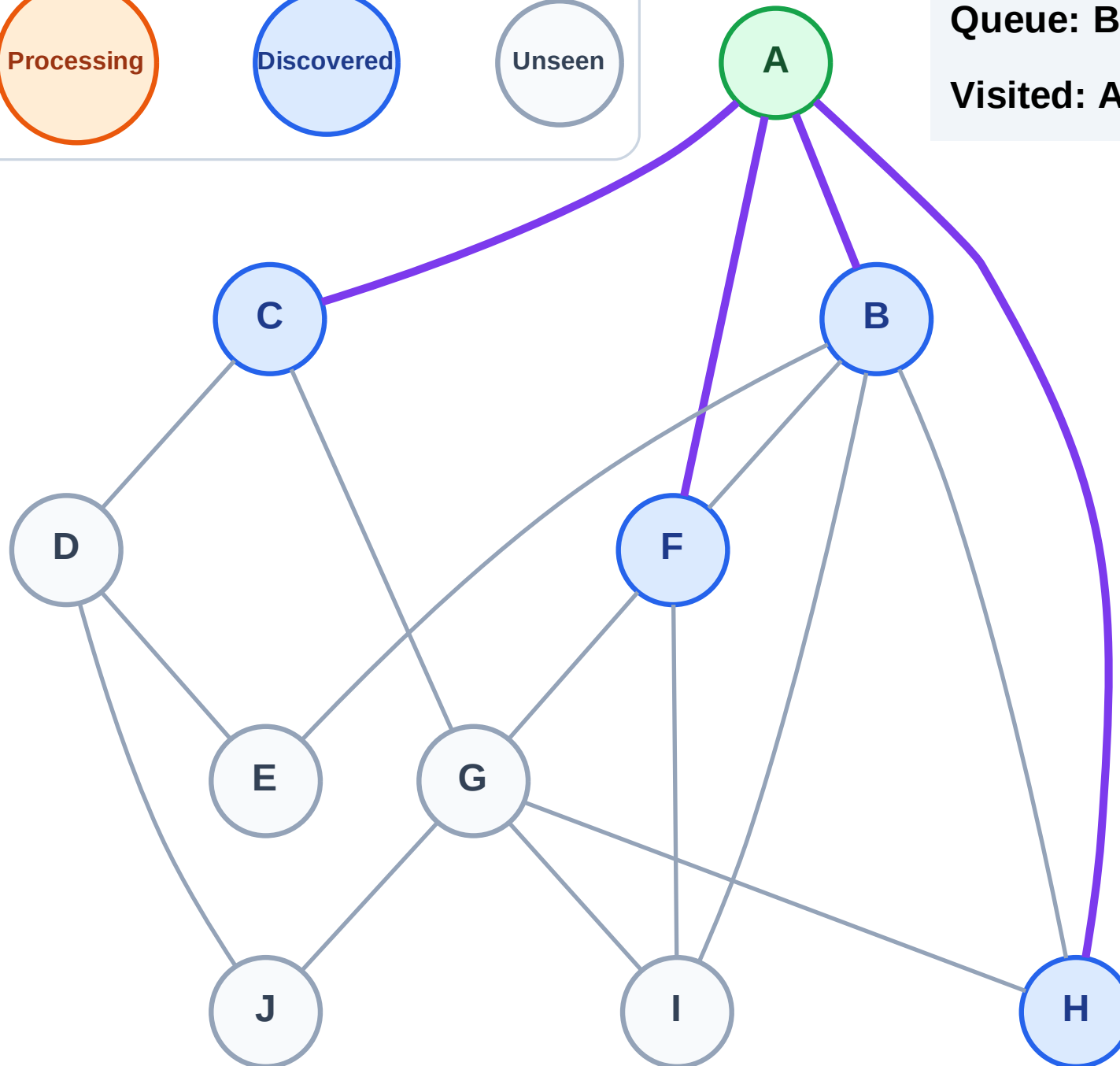
Step 3: Discover B, C, F, H from A

Legend



Queue: B → C → F → H

Visited: A



BFS Traversal

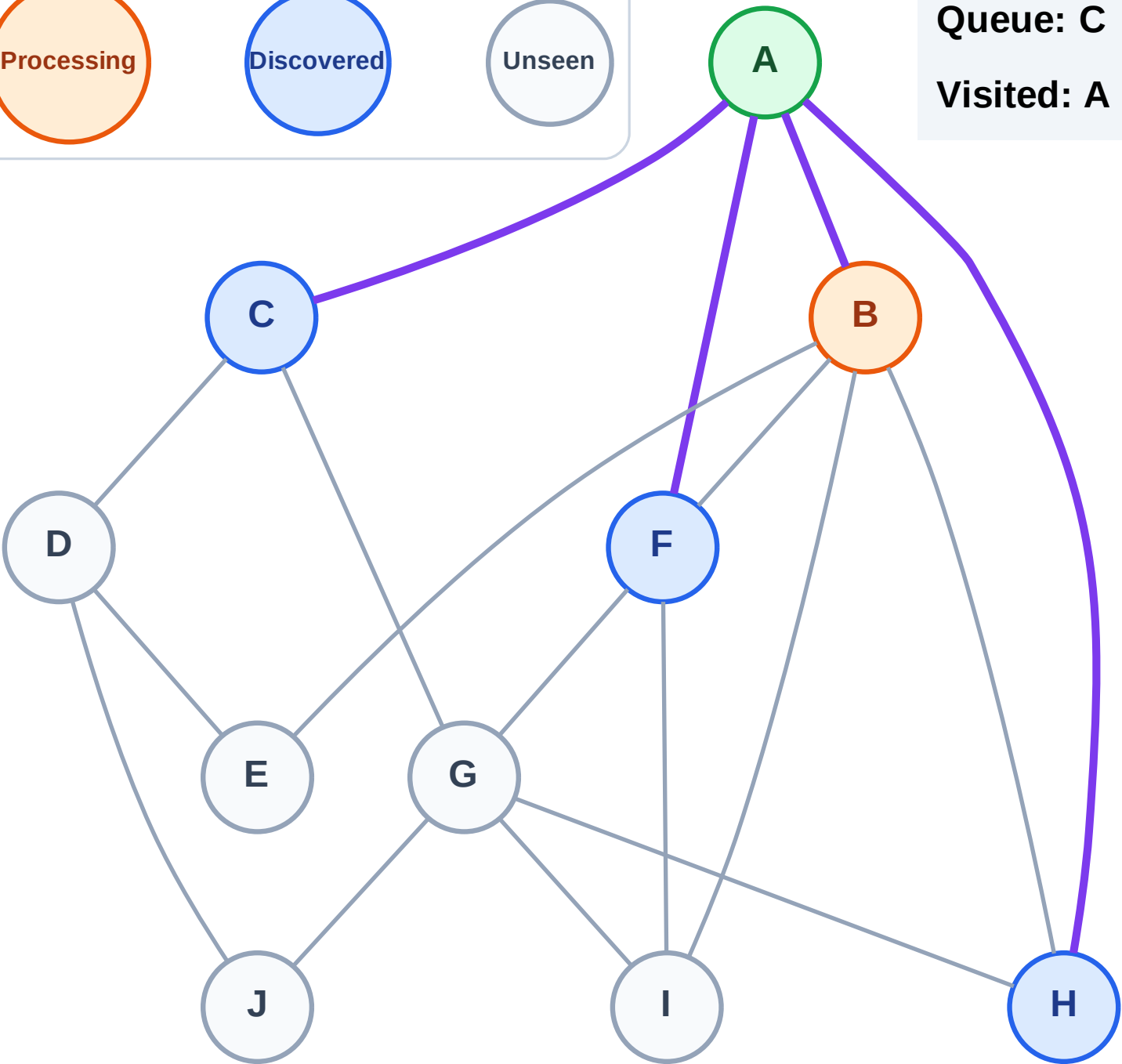
Step 4: Process node B

Legend



Queue: C → F → H

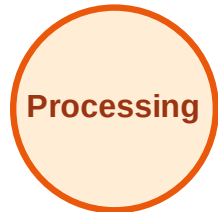
Visited: A



BFS Traversal

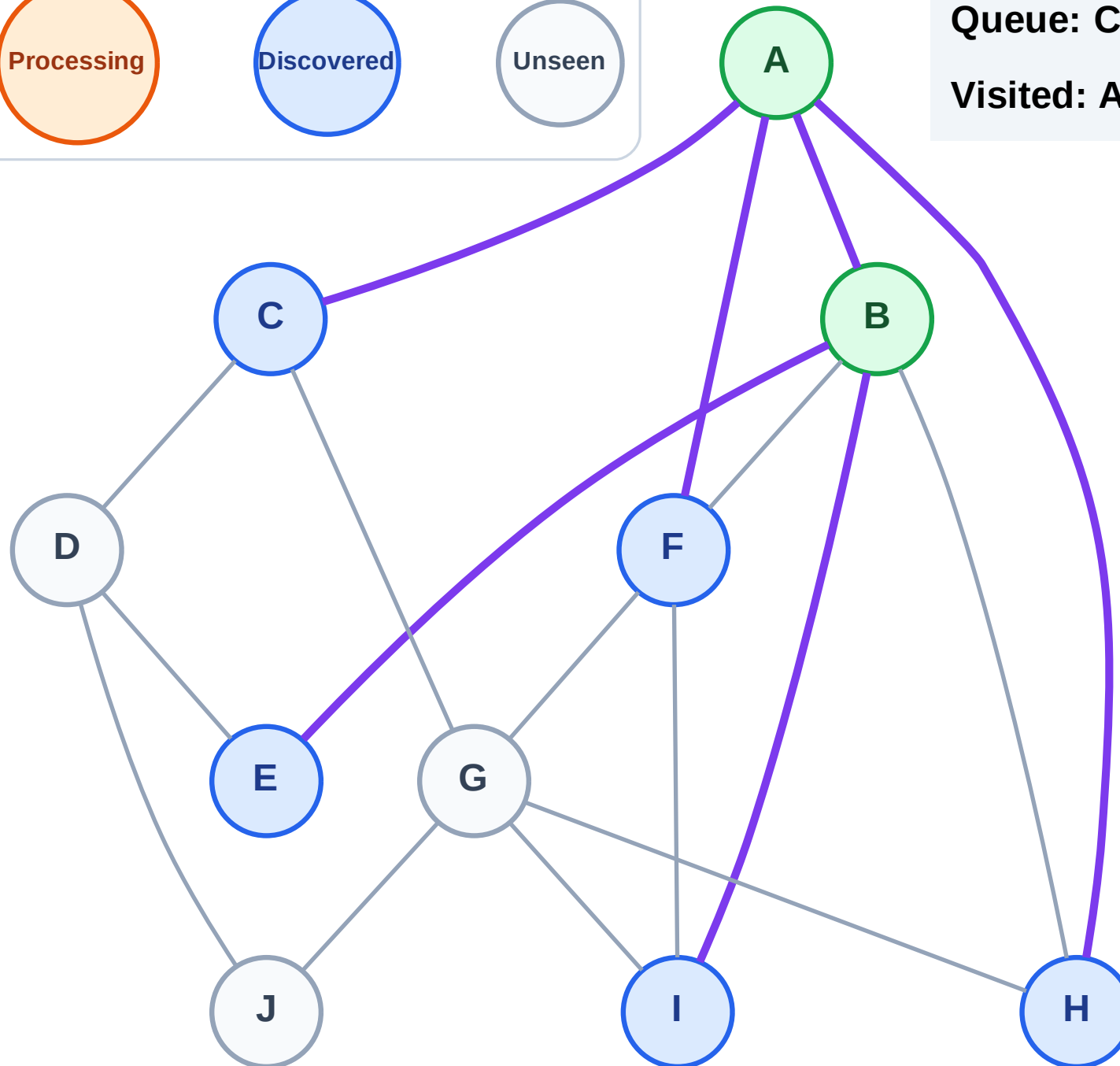
Step 5: Discover E, I from B

Legend



Queue: C → F → H → E → I

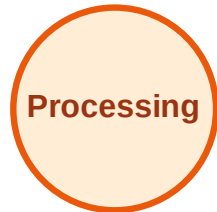
Visited: A, B



BFS Traversal

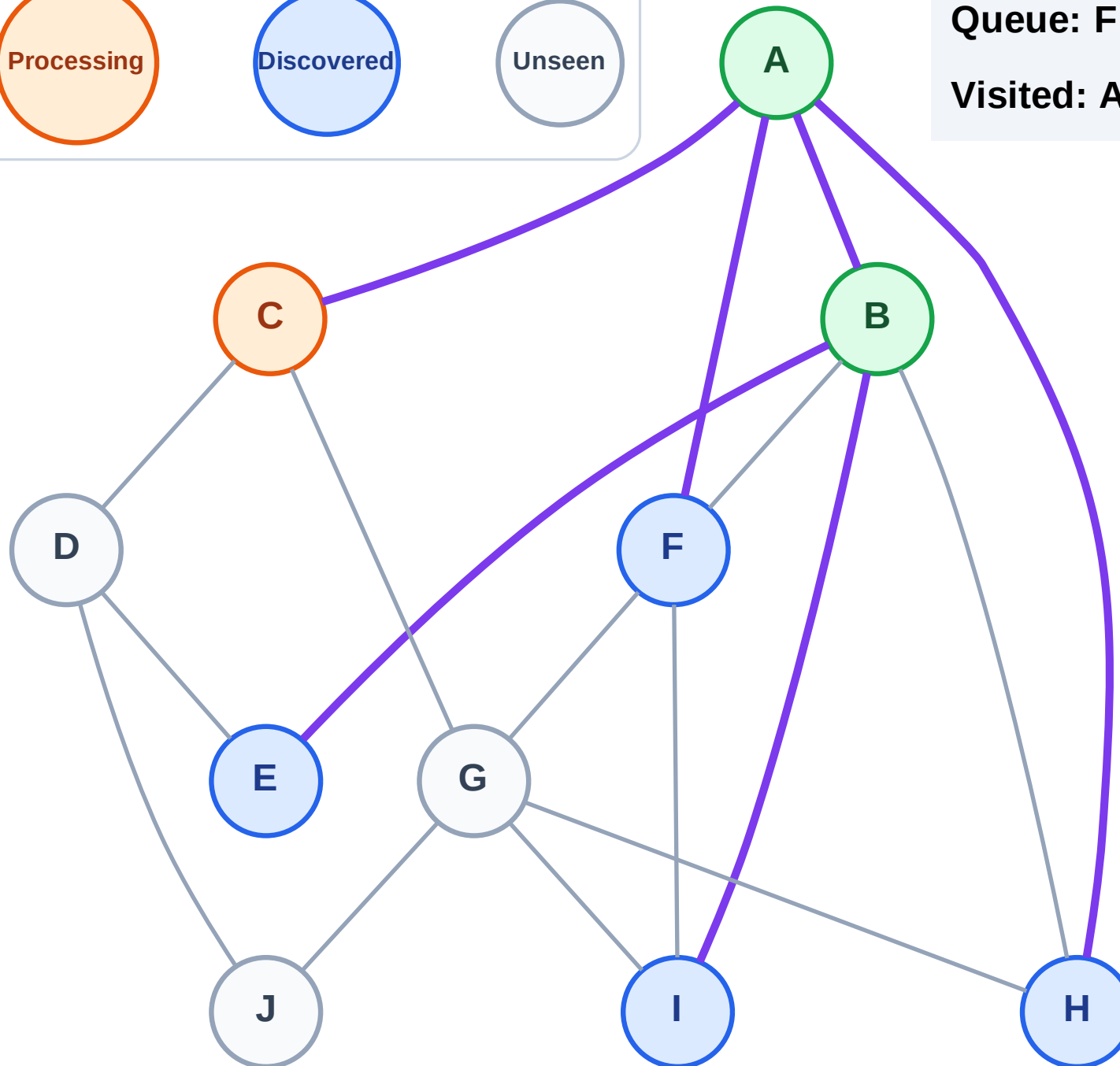
Step 6: Process node C

Legend



Queue: F → H → E → I

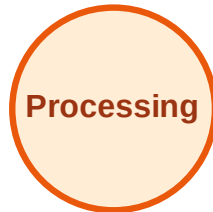
Visited: A, B



BFS Traversal

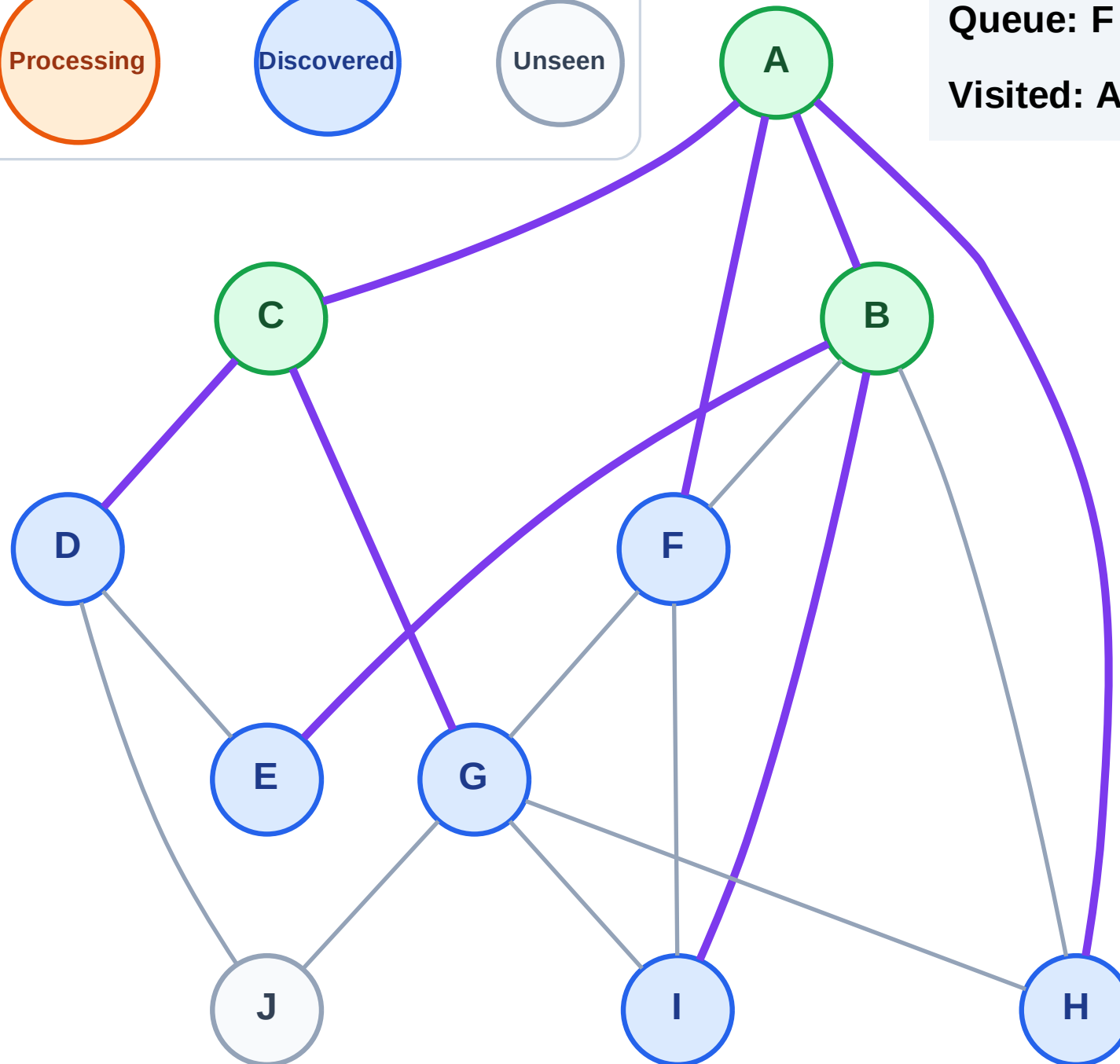
Step 7: Discover D, G from C

Legend



Queue: F → H → E → I → D → G

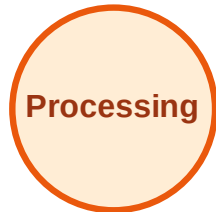
Visited: A, B, C



BFS Traversal

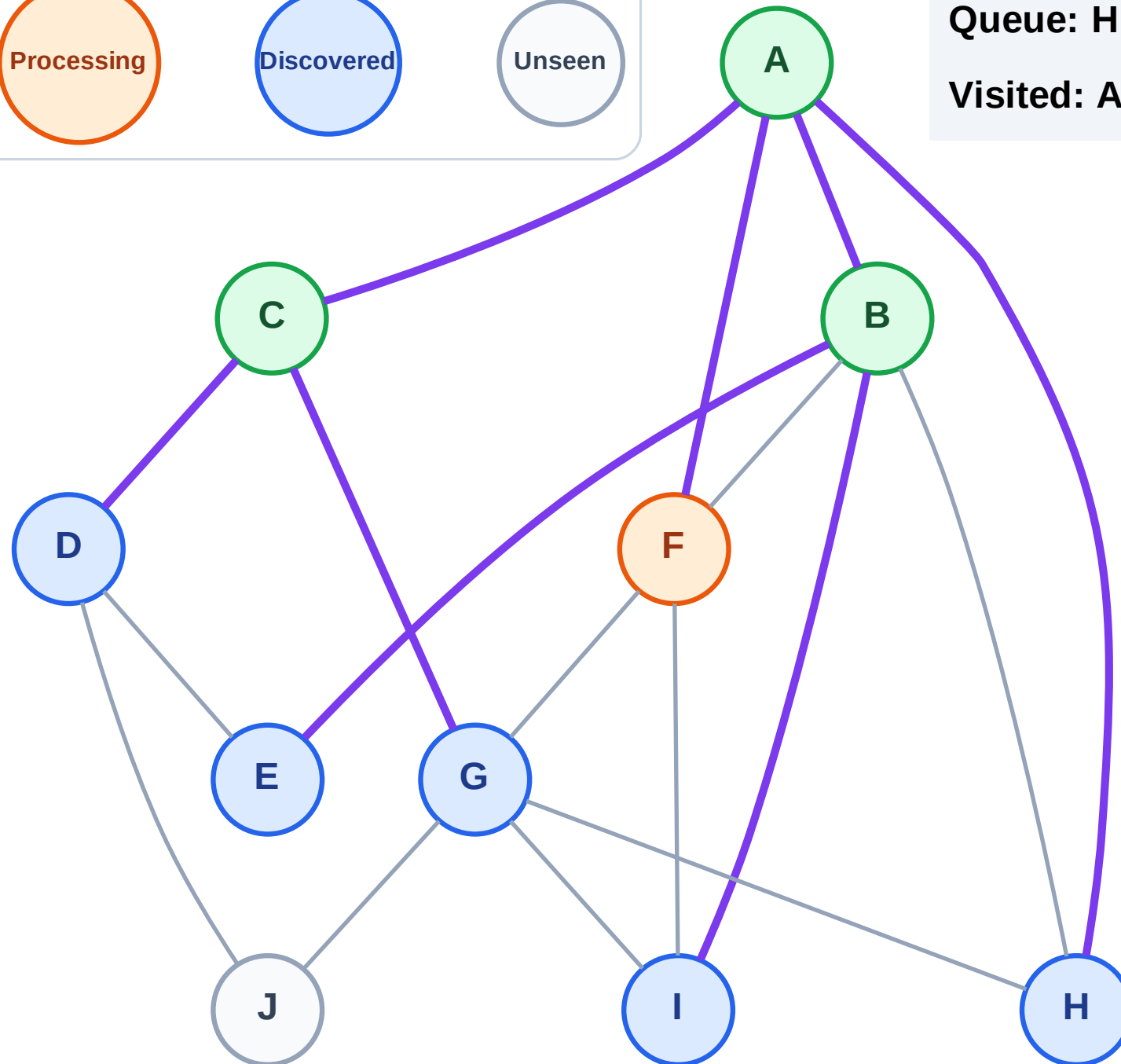
Step 8: Process node F

Legend



Queue: H → E → I → D → G

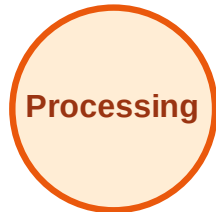
Visited: A, B, C



BFS Traversal

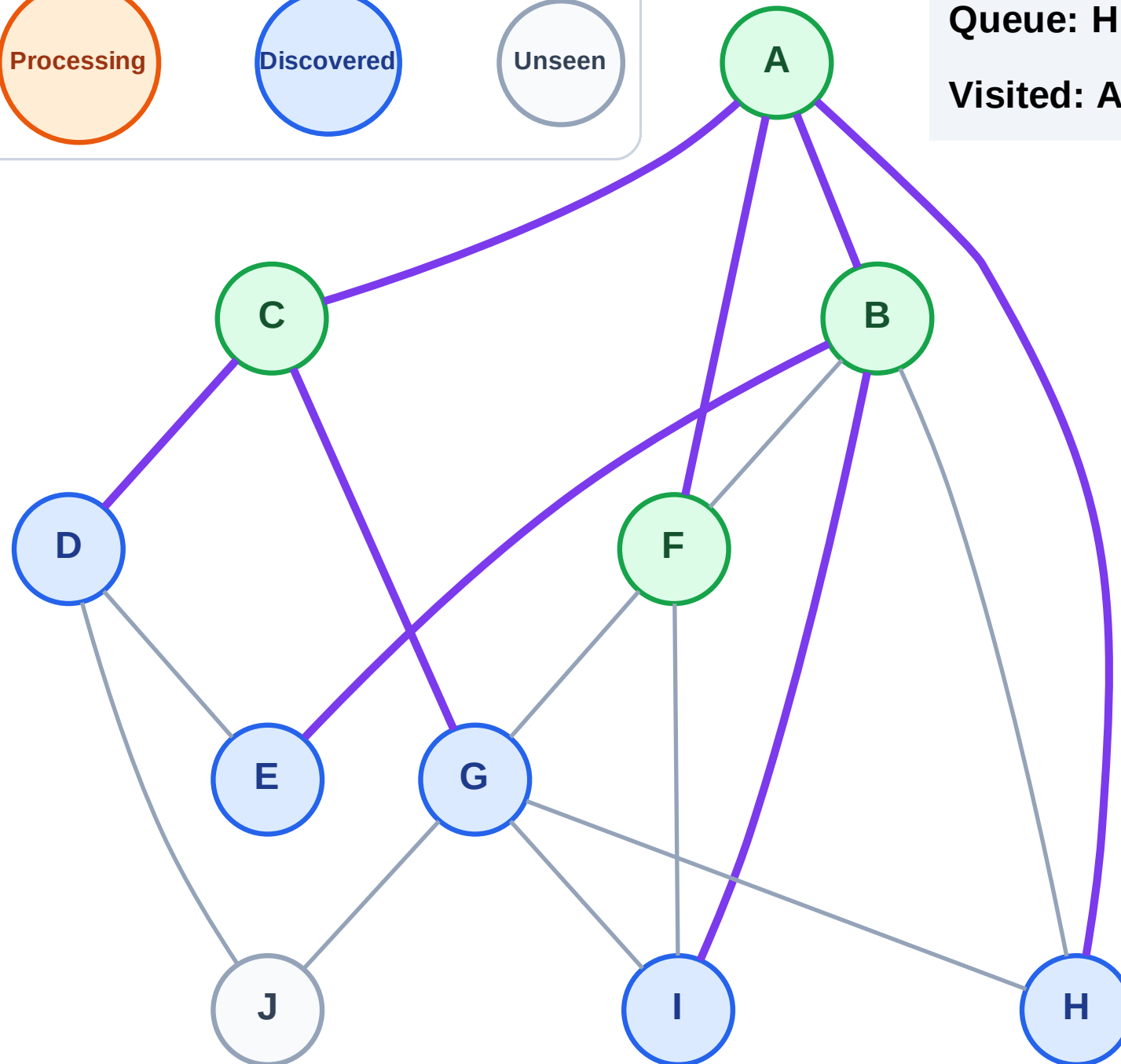
Step 9: No new neighbors from F

Legend



Queue: H → E → I → D → G

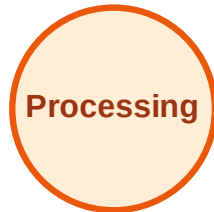
Visited: A, B, C, F



BFS Traversal

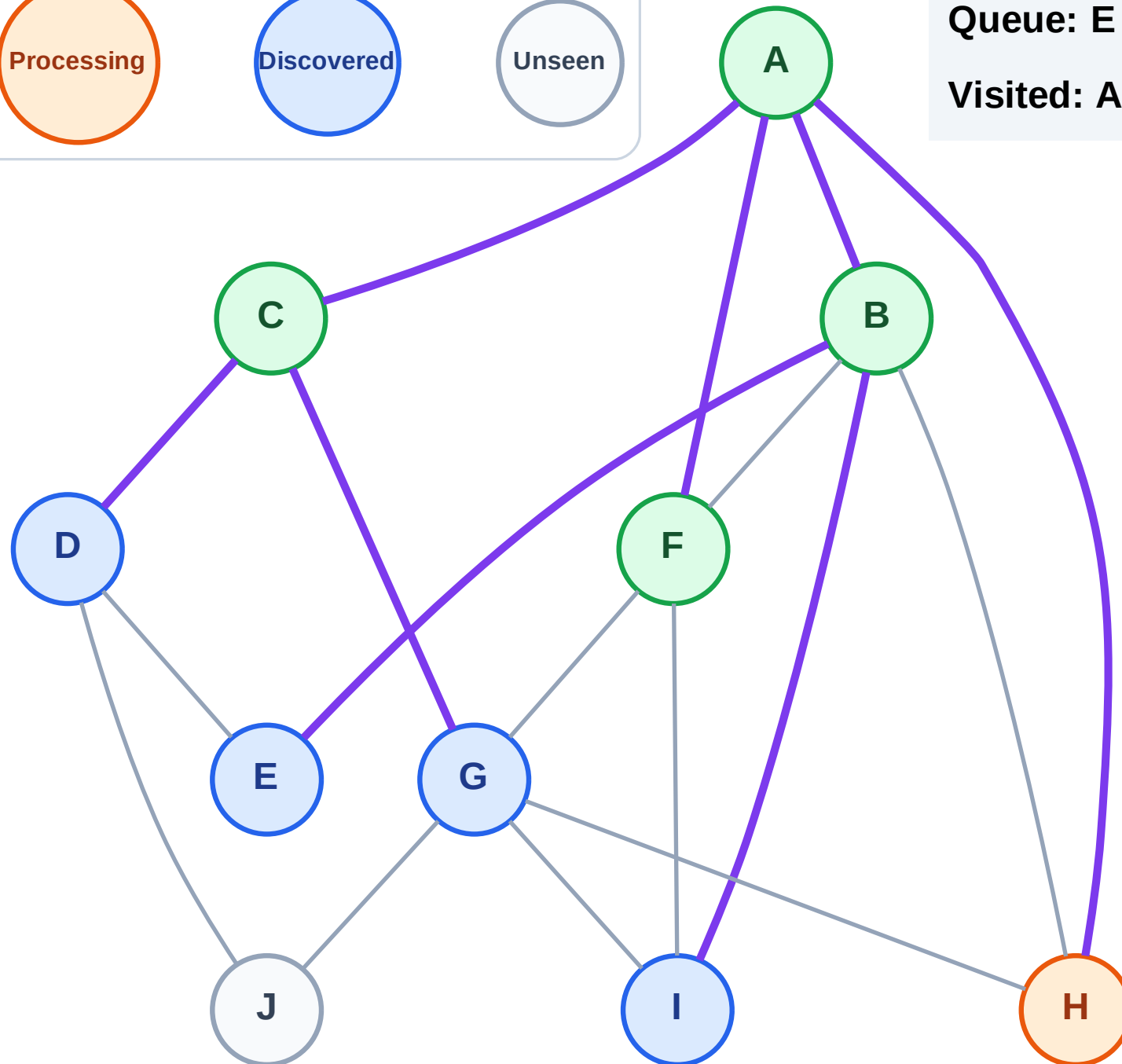
Step 10: Process node H

Legend



Queue: E → I → D → G

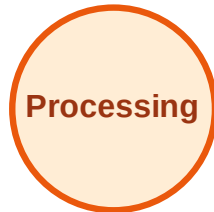
Visited: A, B, C, F



BFS Traversal

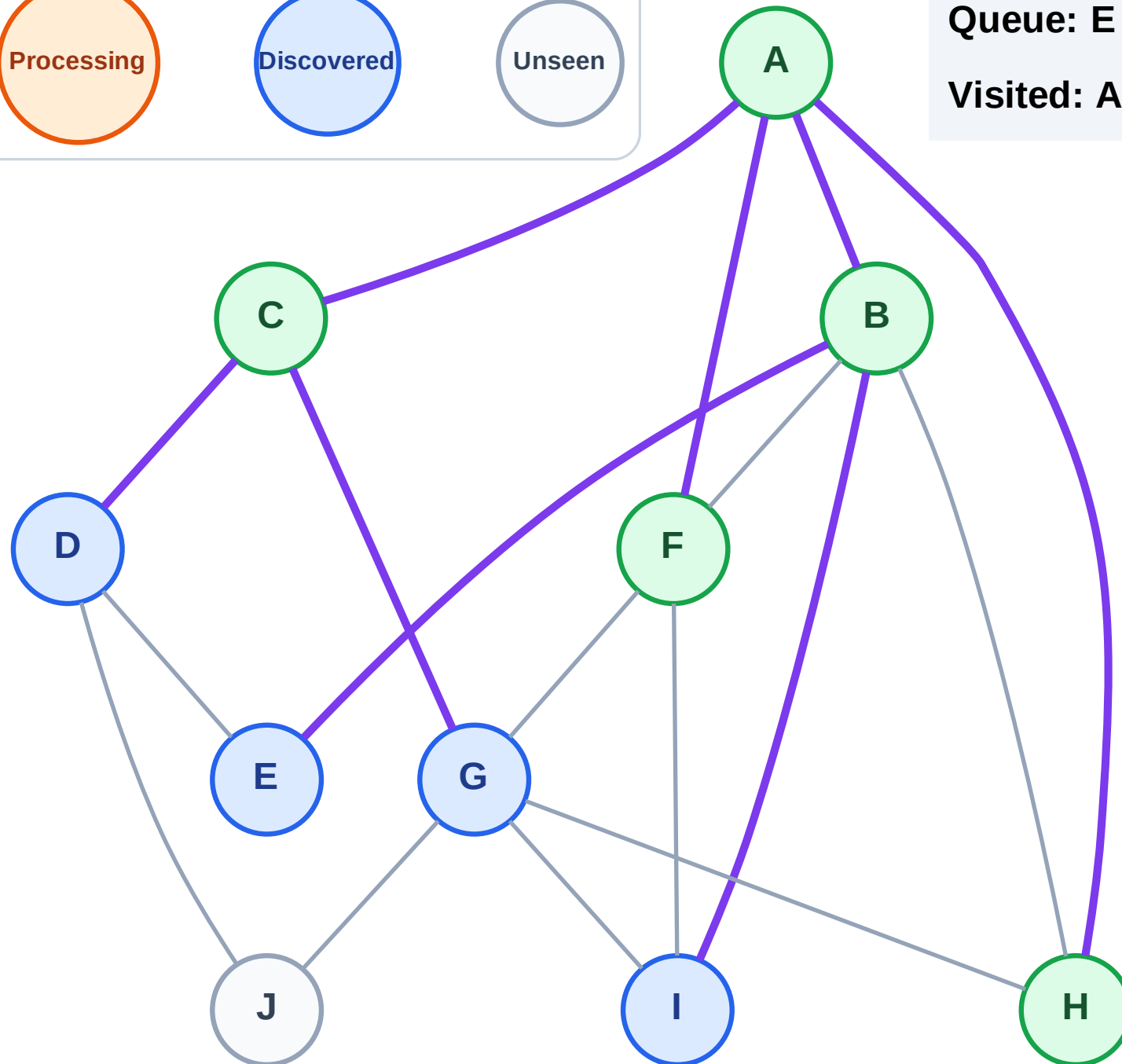
Step 11: No new neighbors from H

Legend



Queue: E → I → D → G

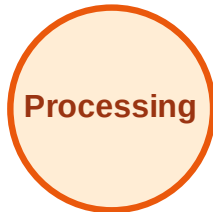
Visited: A, B, C, F, H



BFS Traversal

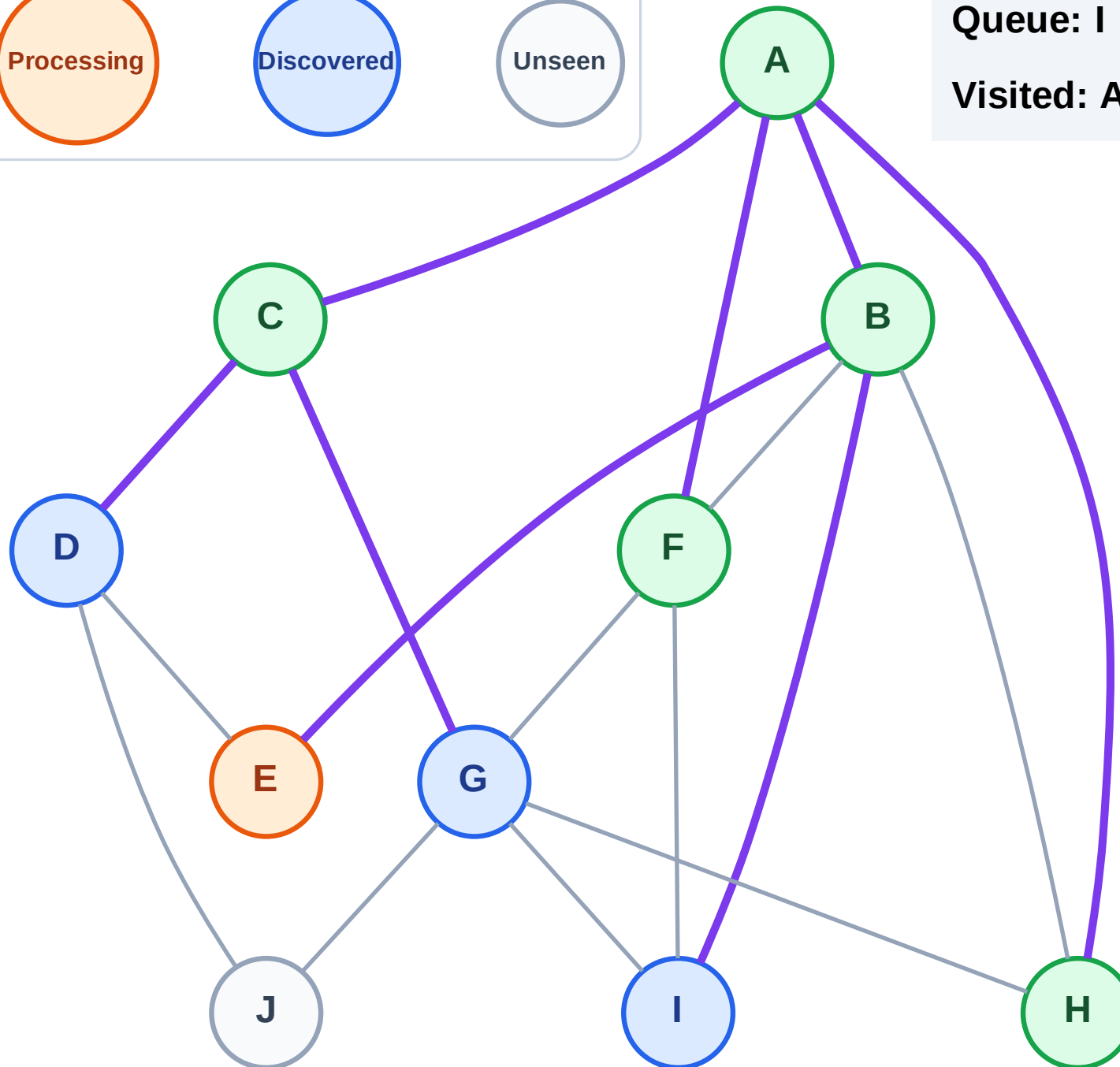
Step 12: Process node E

Legend



Queue: I → D → G

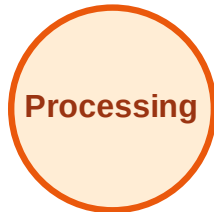
Visited: A, B, C, F, H



BFS Traversal

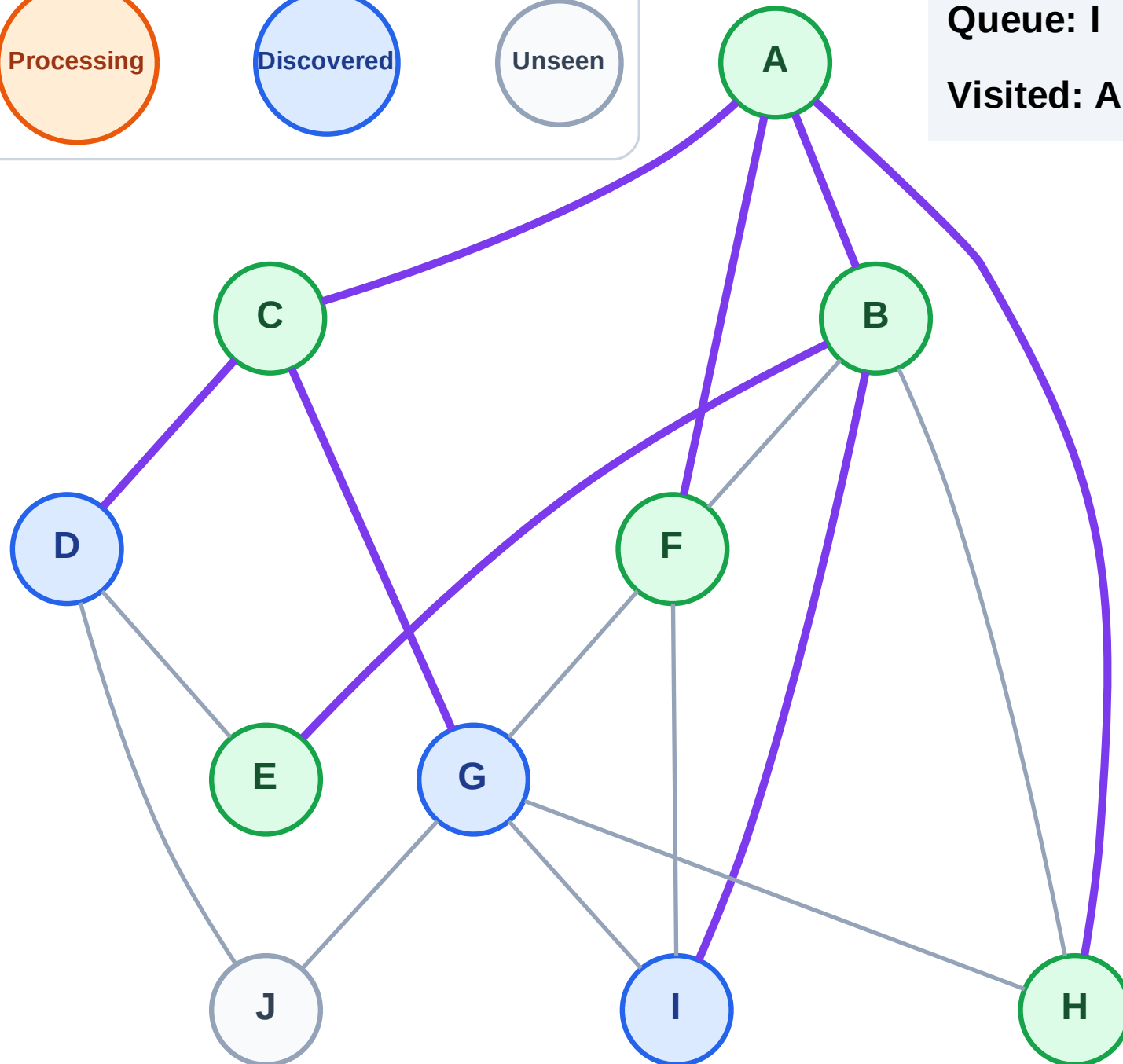
Step 13: No new neighbors from E

Legend



Queue: I → D → G

Visited: A, B, C, E, F, H



BFS Traversal

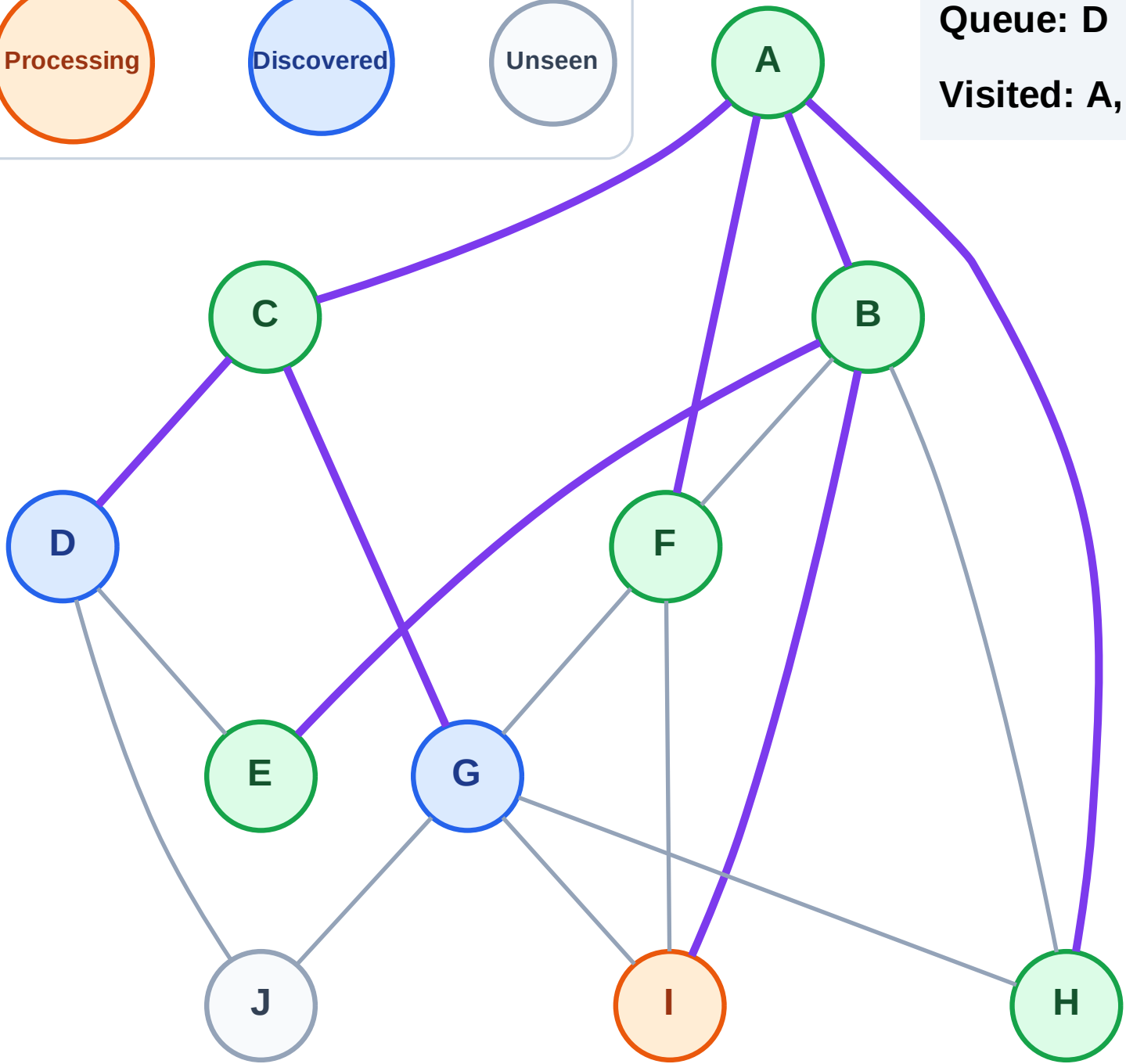
Step 14: Process node I

Legend



Queue: D → G

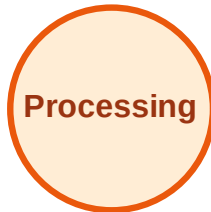
Visited: A, B, C, E, F, H



BFS Traversal

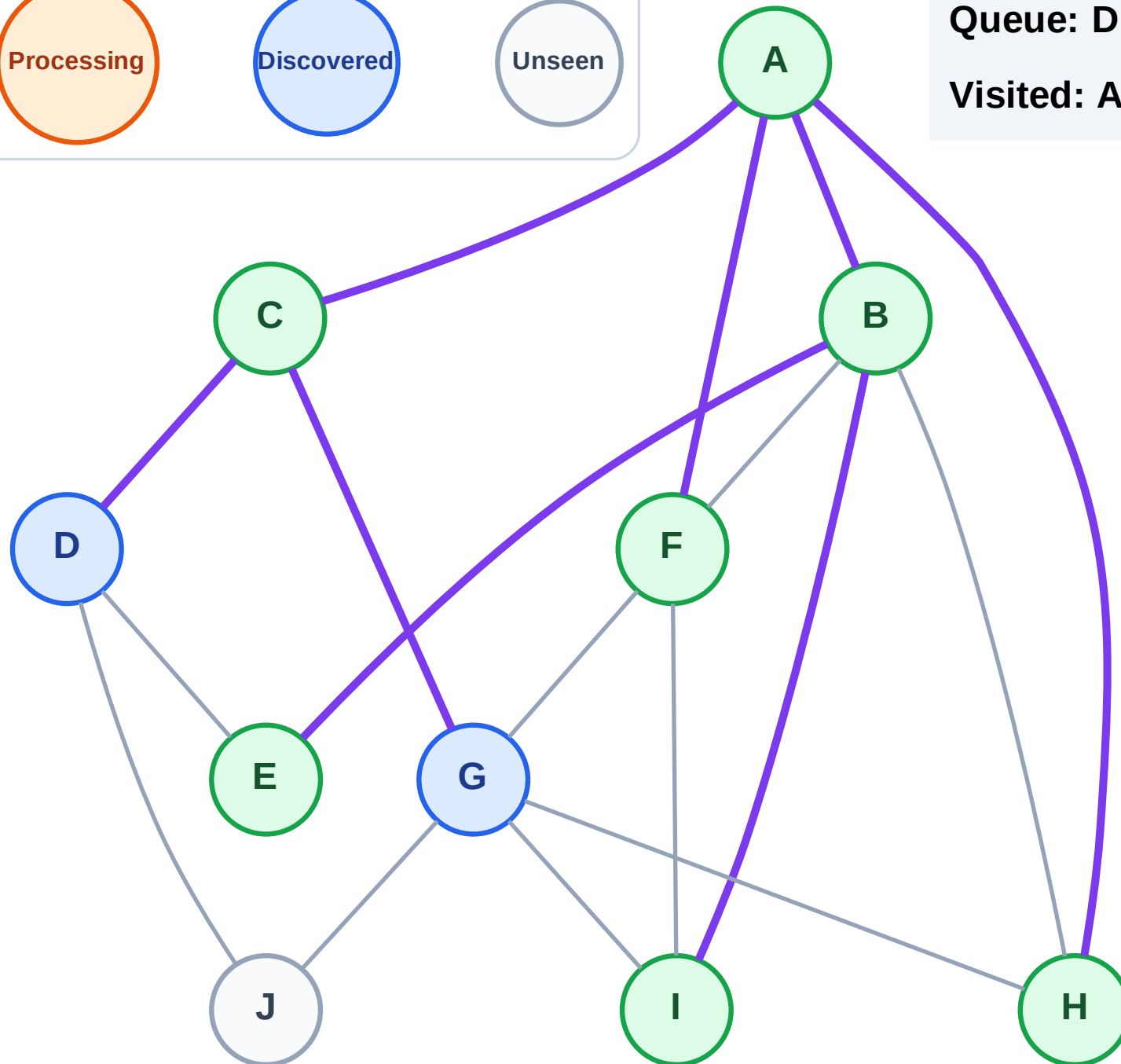
Step 15: No new neighbors from I

Legend



Queue: D → G

Visited: A, B, C, E, F, H, I



BFS Traversal

Step 16: Process node D

Legend

Complete

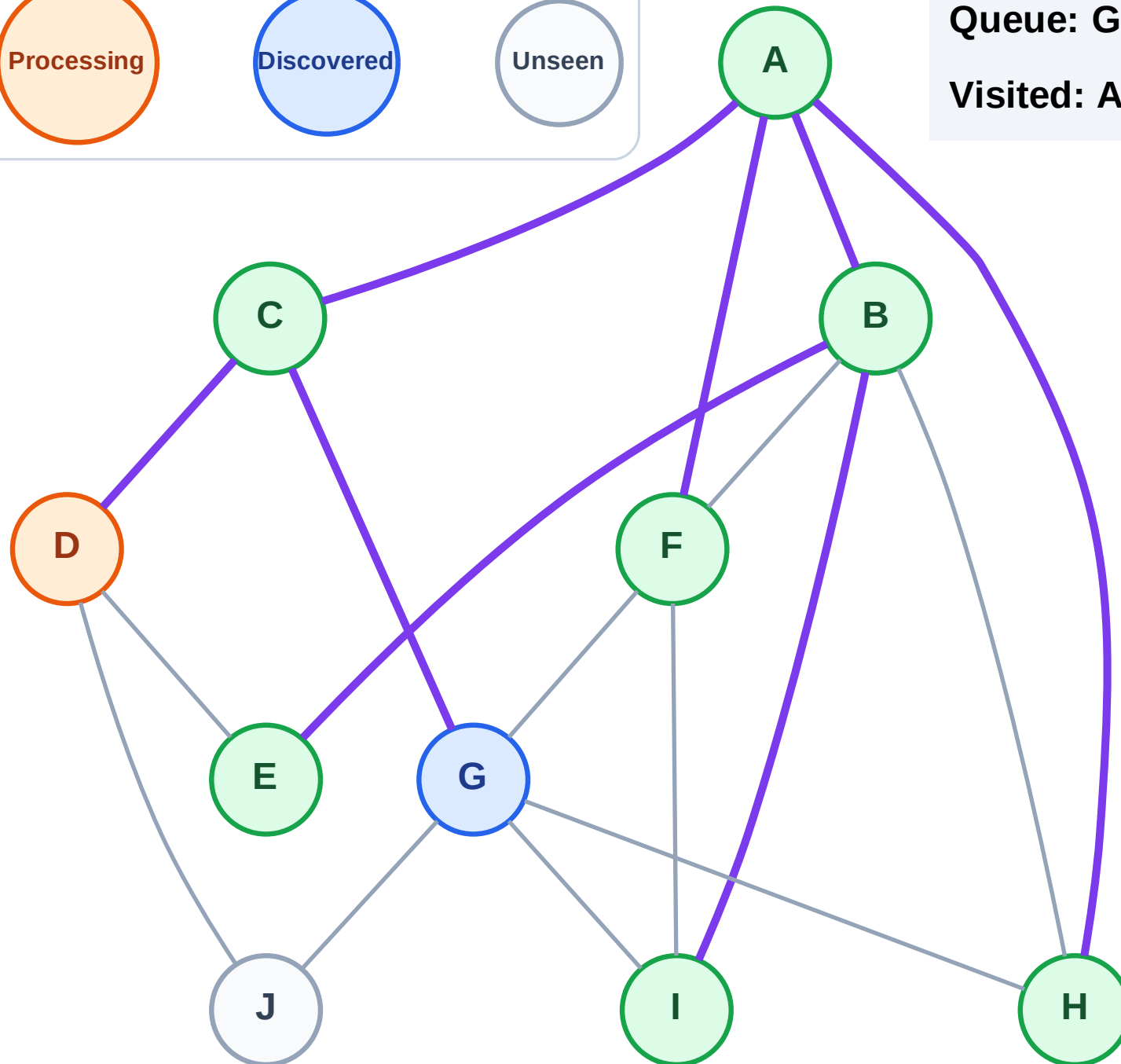
Processing

Discovered

Unseen

Queue: G

Visited: A, B, C, E, F, H, I



BFS Traversal

Step 17: Discover J from D

Legend

Complete

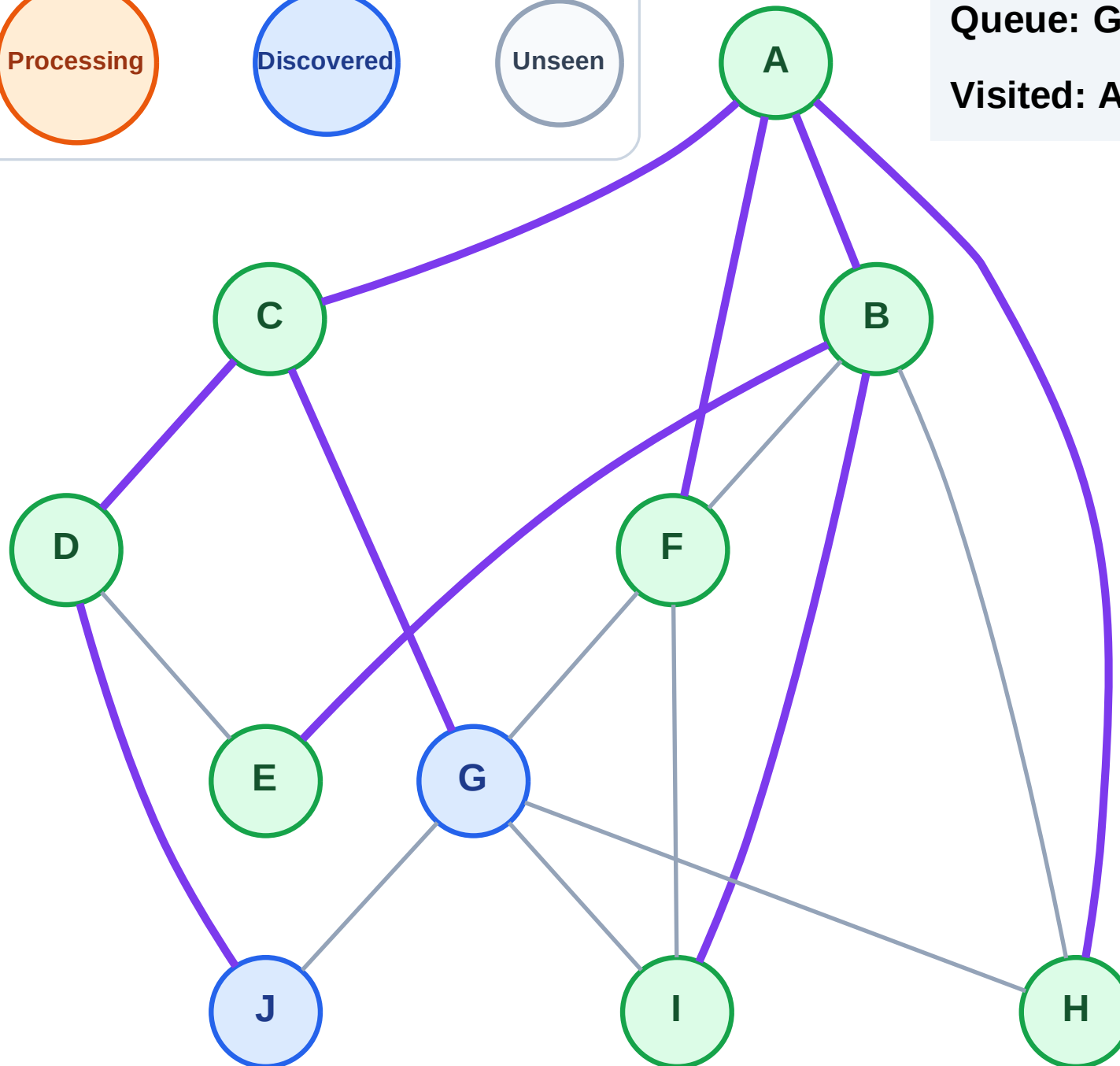
Processing

Discovered

Unseen

Queue: G → J

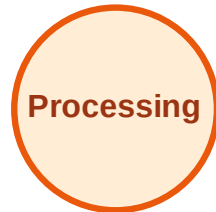
Visited: A, B, C, D, E, F, H, I



BFS Traversal

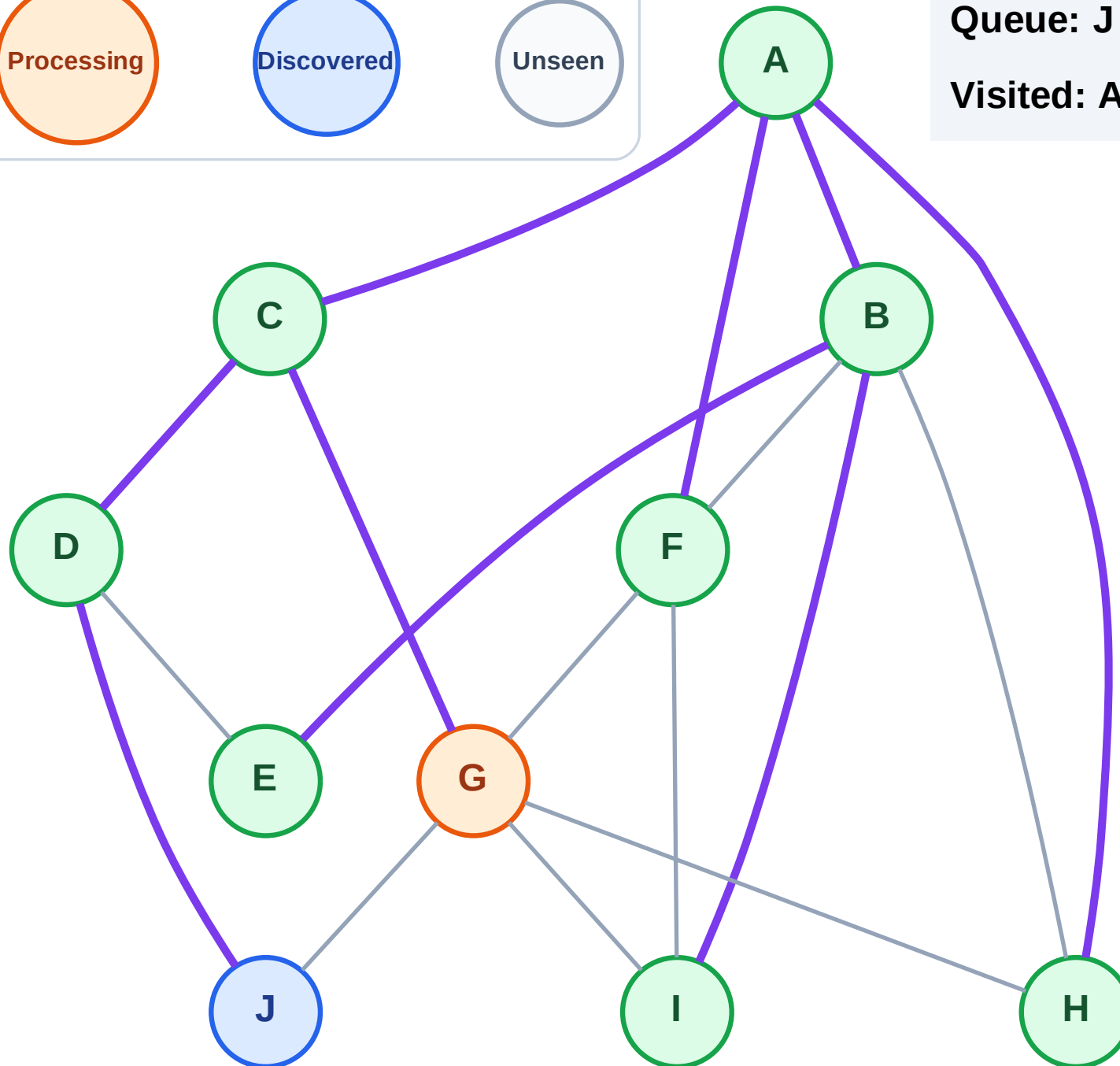
Step 18: Process node G

Legend



Queue: J

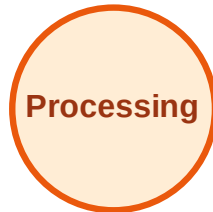
Visited: A, B, C, D, E, F, H, I



BFS Traversal

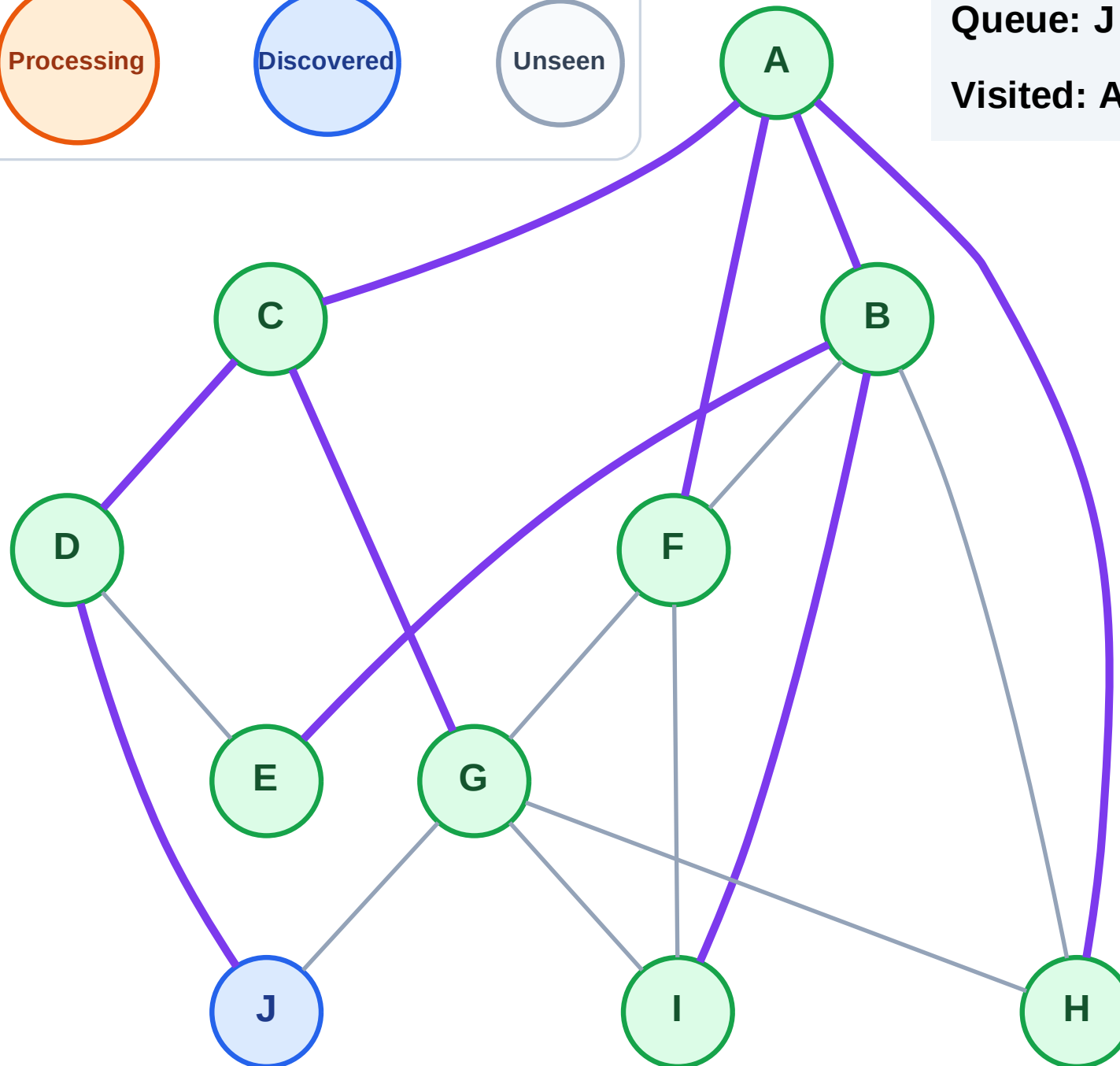
Step 19: No new neighbors from G

Legend



Queue: J

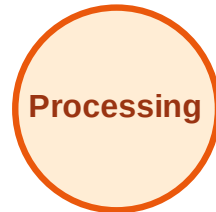
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

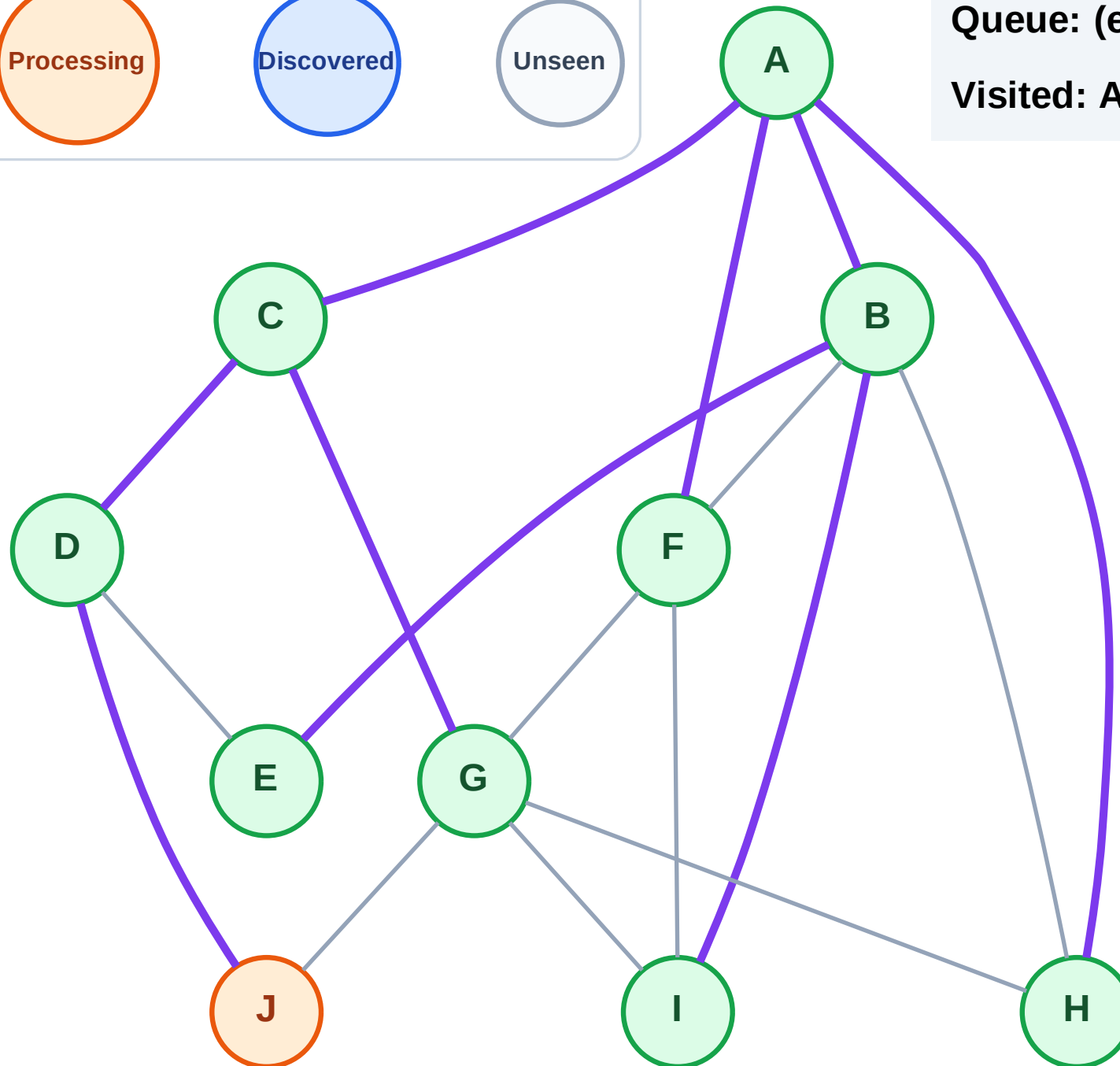
Step 20: Process node J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

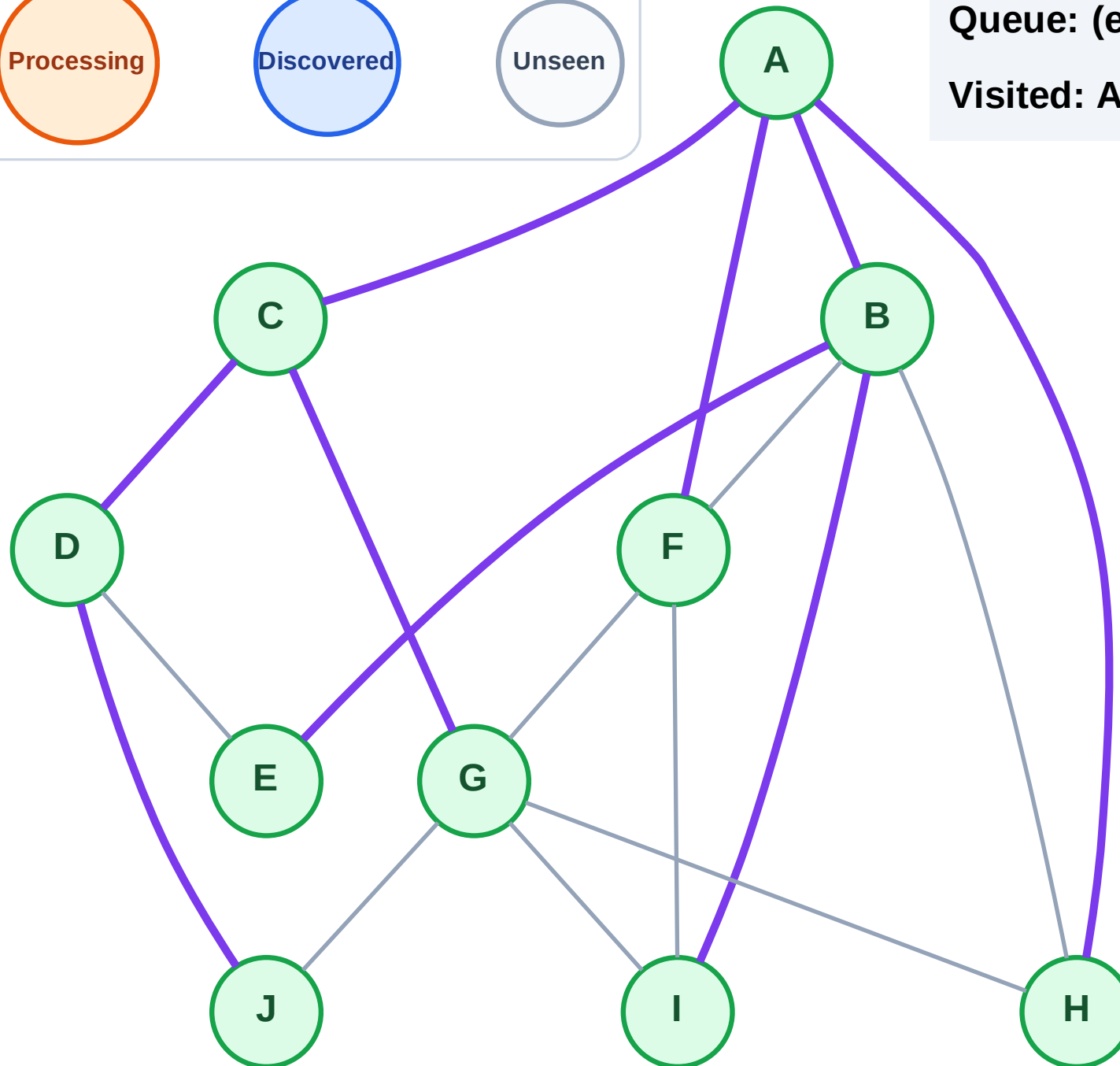
Step 21: No new neighbors from J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

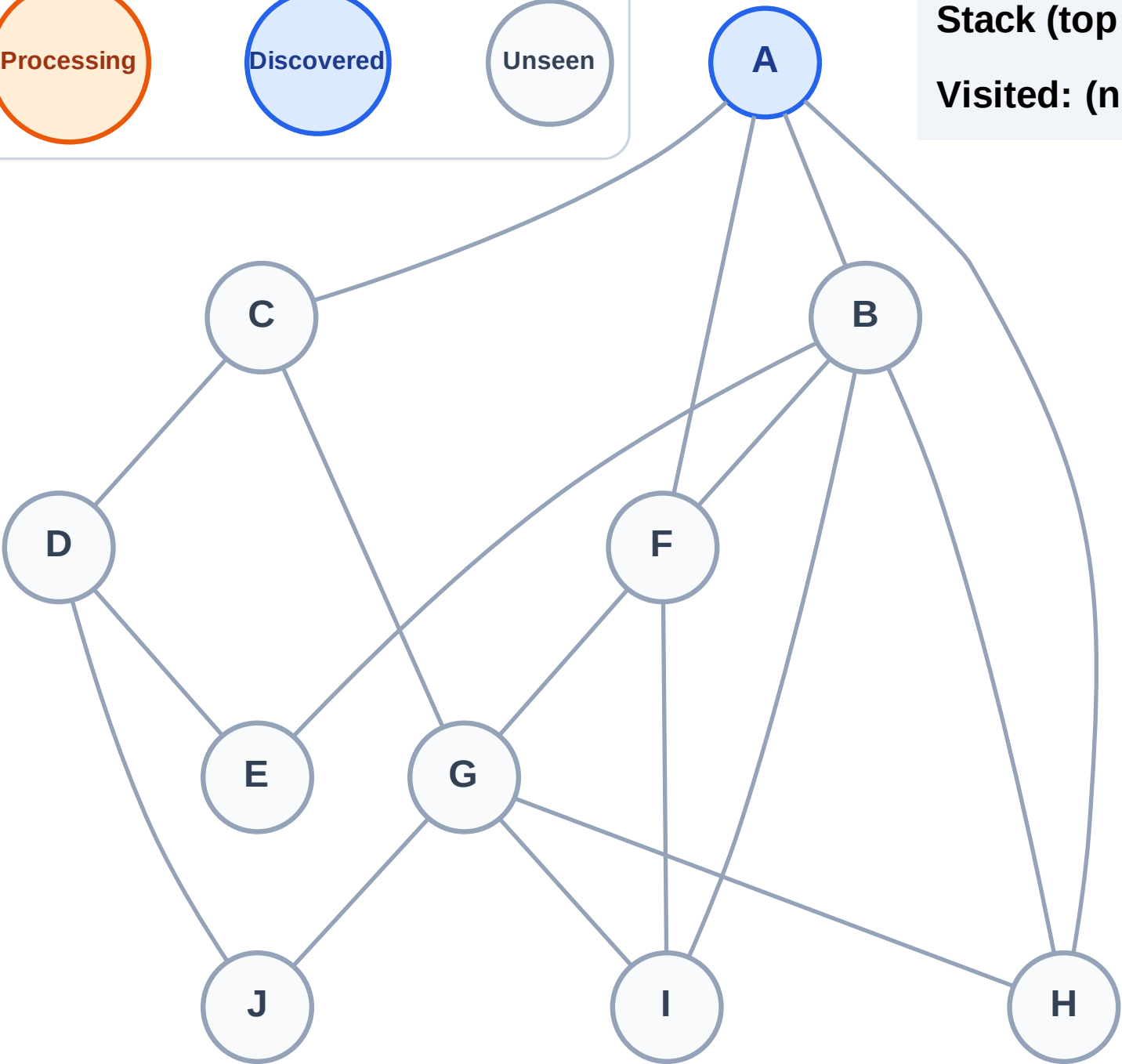
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

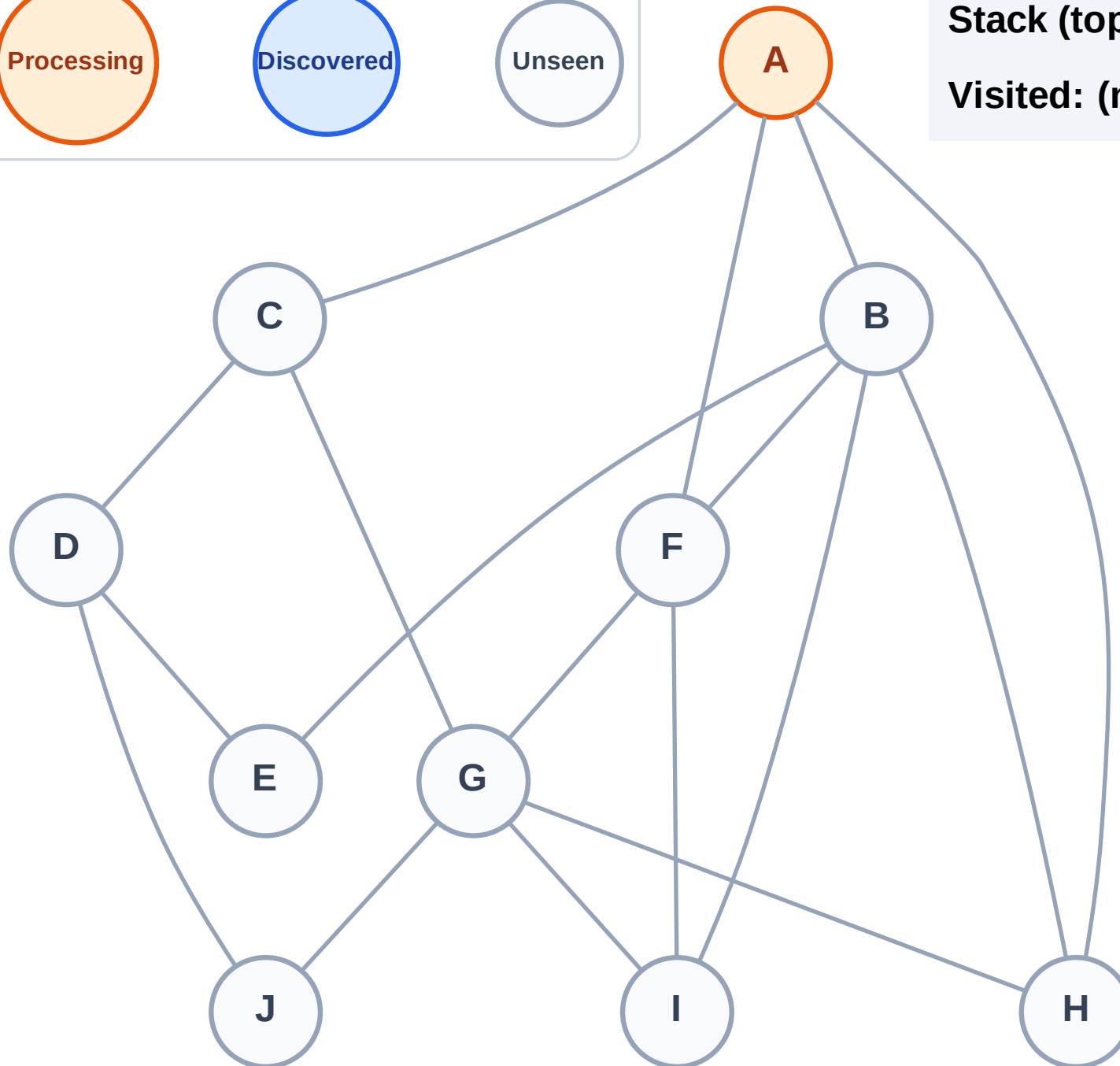
Processing

Discovered

Unseen

Stack (top at right): (empty)

Visited: (none)



DFS Traversal

Step 3: Discover H, F, C, B from A

Legend

Complete

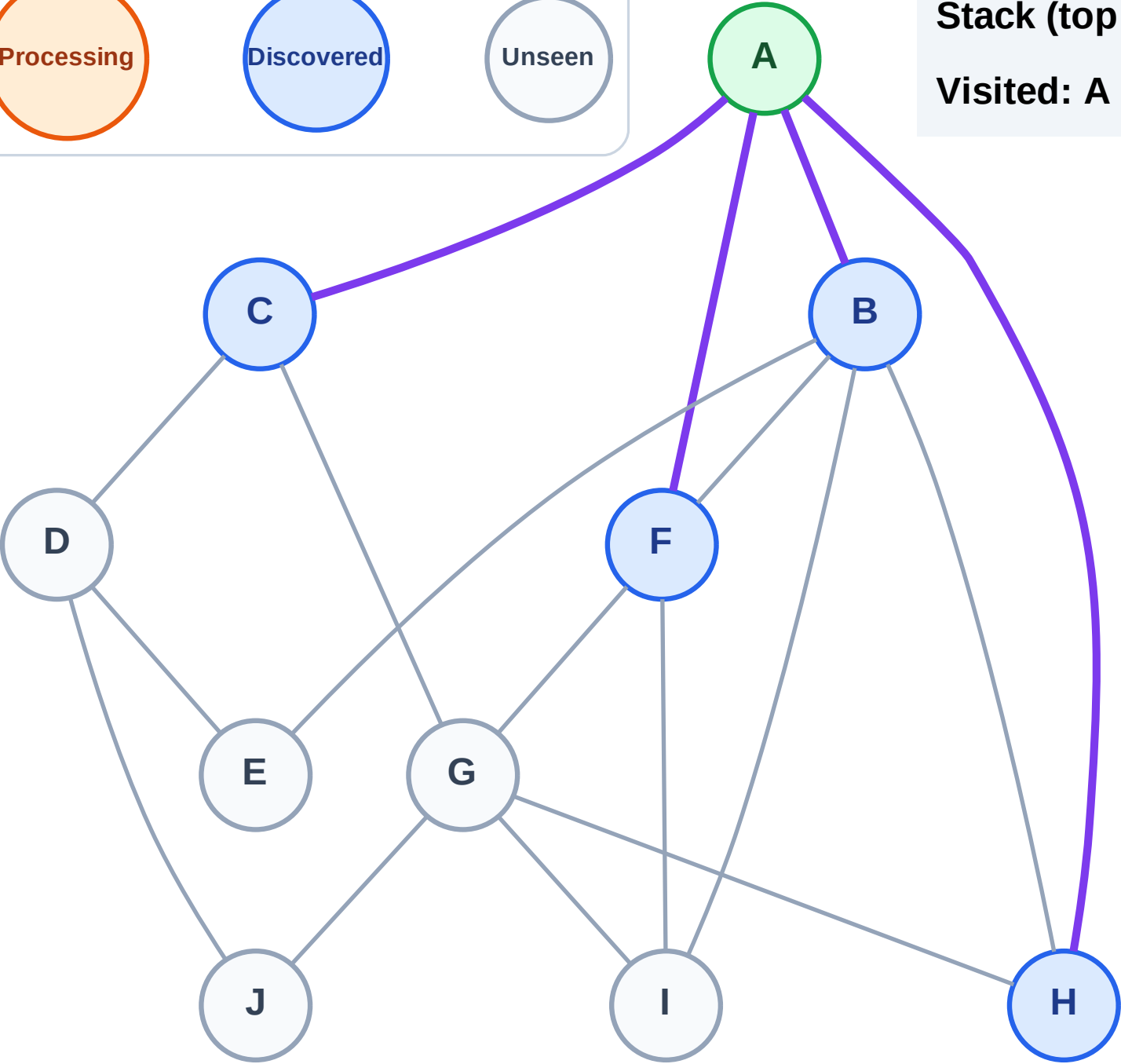
Processing

Discovered

Unseen

Stack (top at right): H | F | C | B

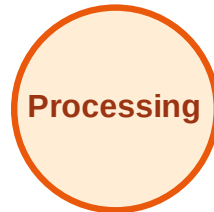
Visited: A



DFS Traversal

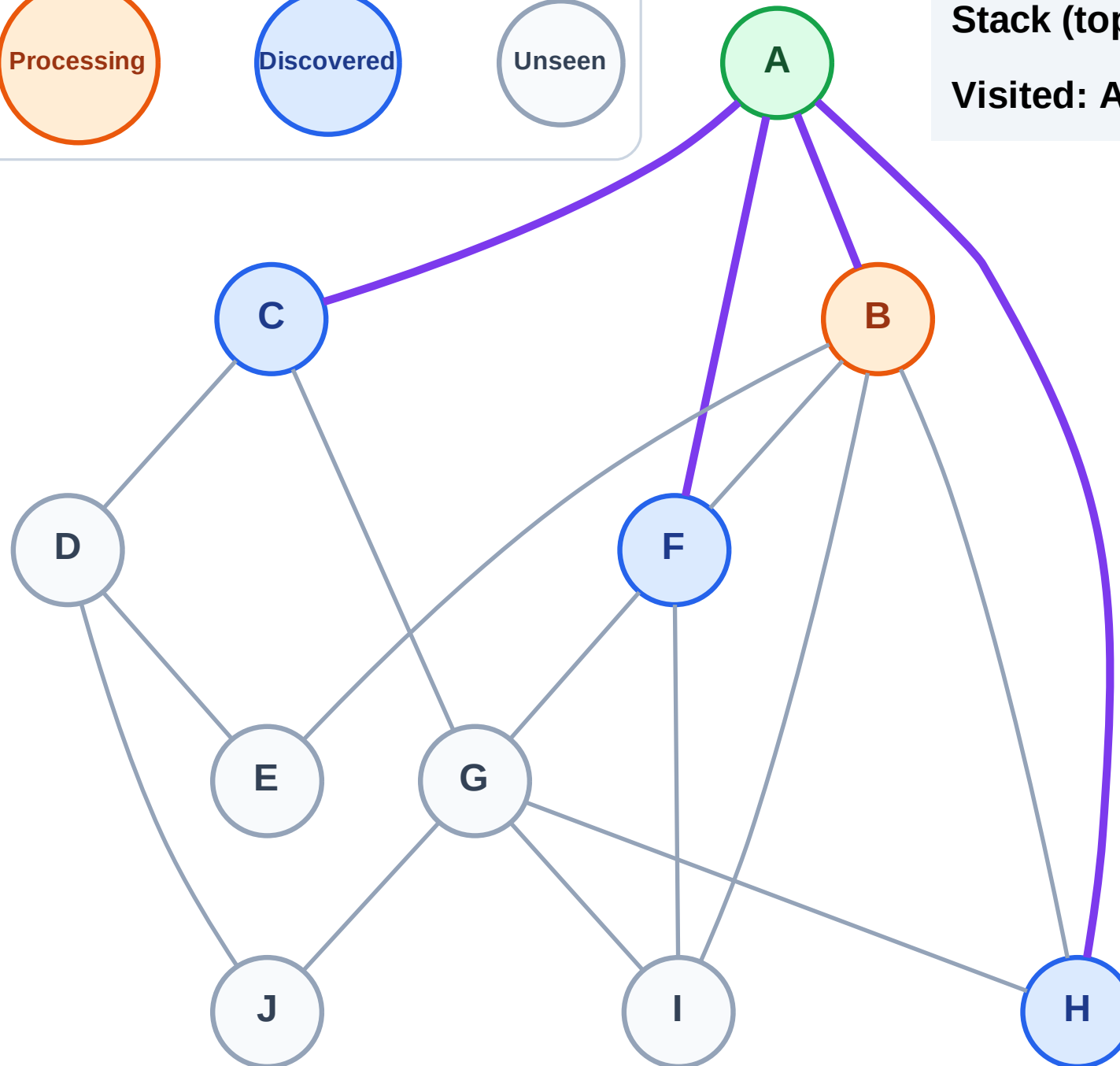
Step 4: Process node B

Legend



Stack (top at right): H | F | C

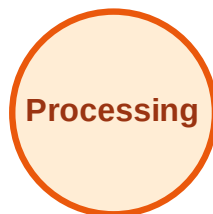
Visited: A



DFS Traversal

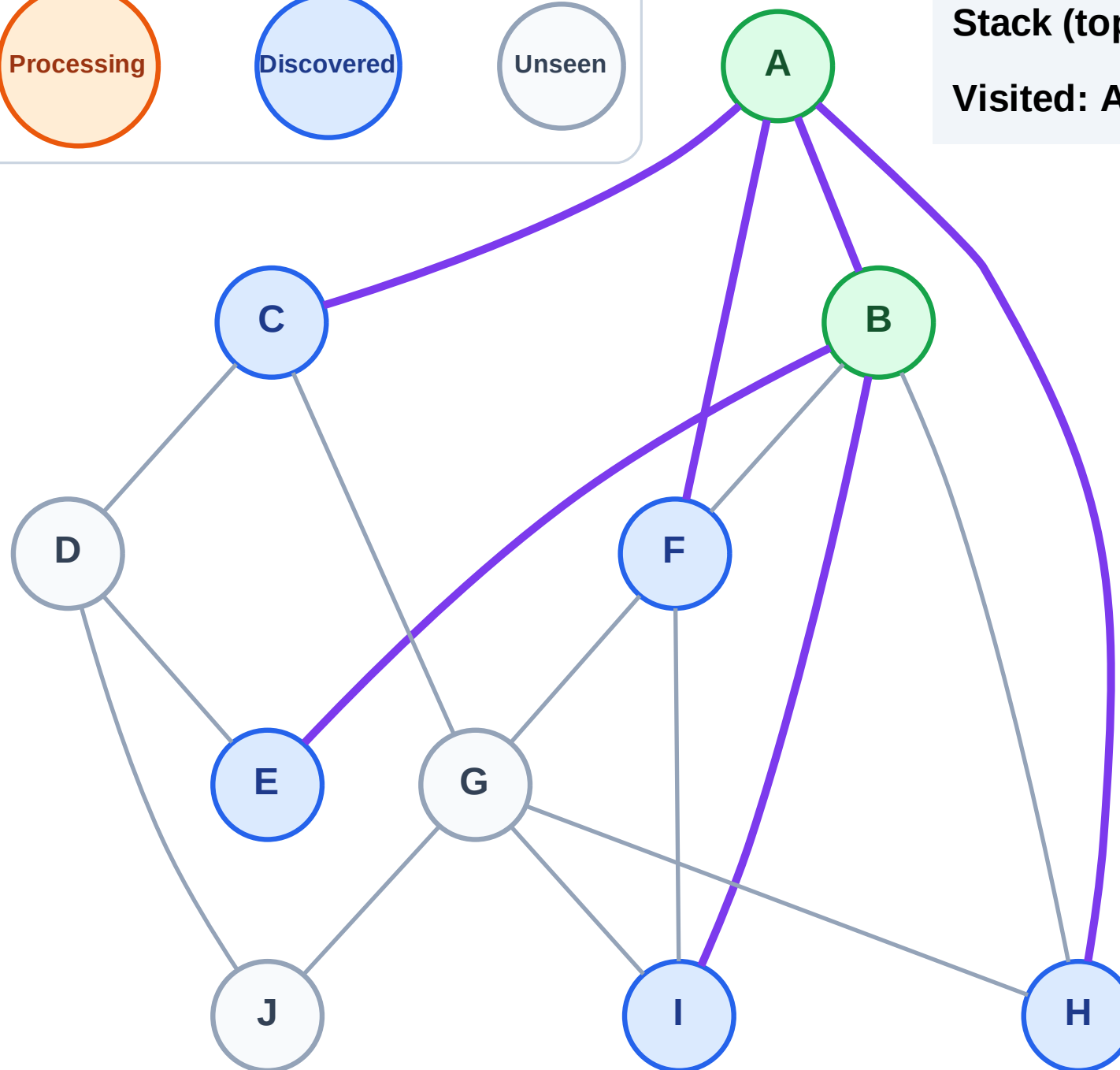
Step 5: Discover I, E from B

Legend



Stack (top at right): H | F | C | I | E

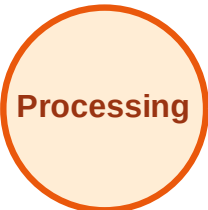
Visited: A, B



DFS Traversal

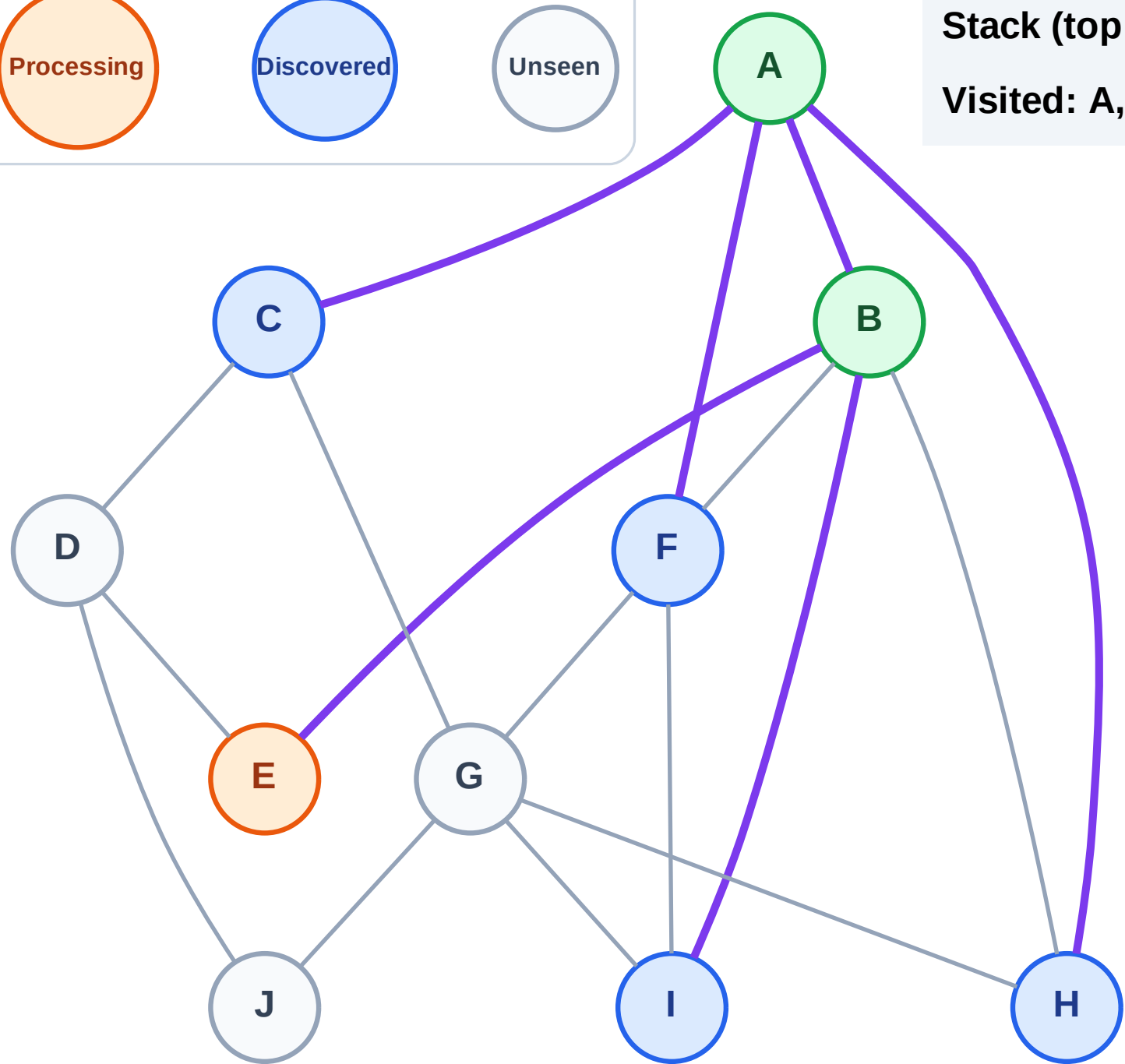
Step 6: Process node E

Legend



Stack (top at right): H | F | C | I

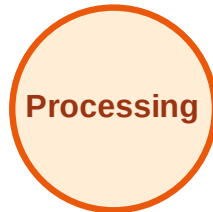
Visited: A, B



DFS Traversal

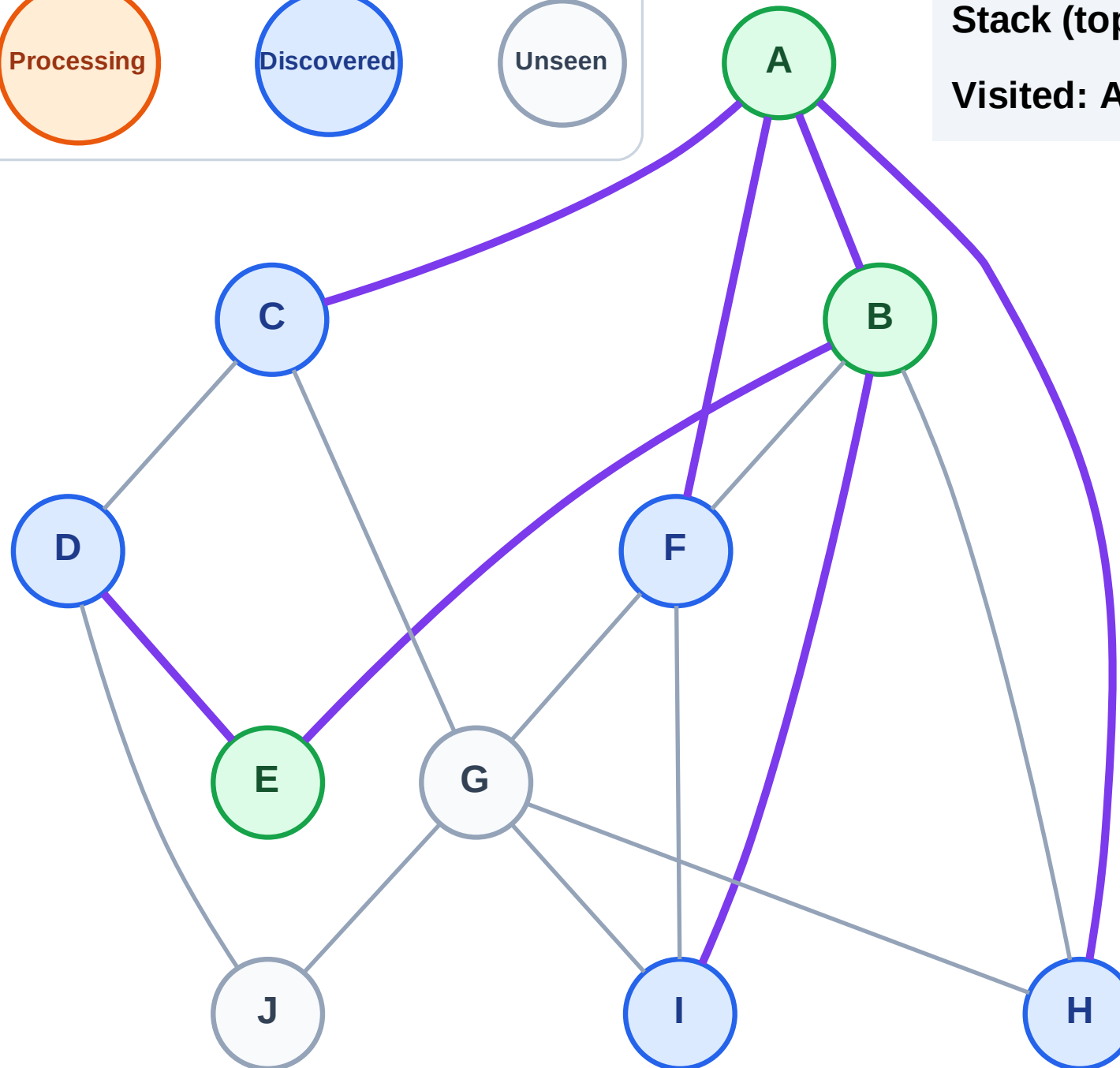
Step 7: Discover D from E

Legend



Stack (top at right): H | F | C | I | D

Visited: A, B, E



DFS Traversal

Step 8: Process node D

Legend

Complete

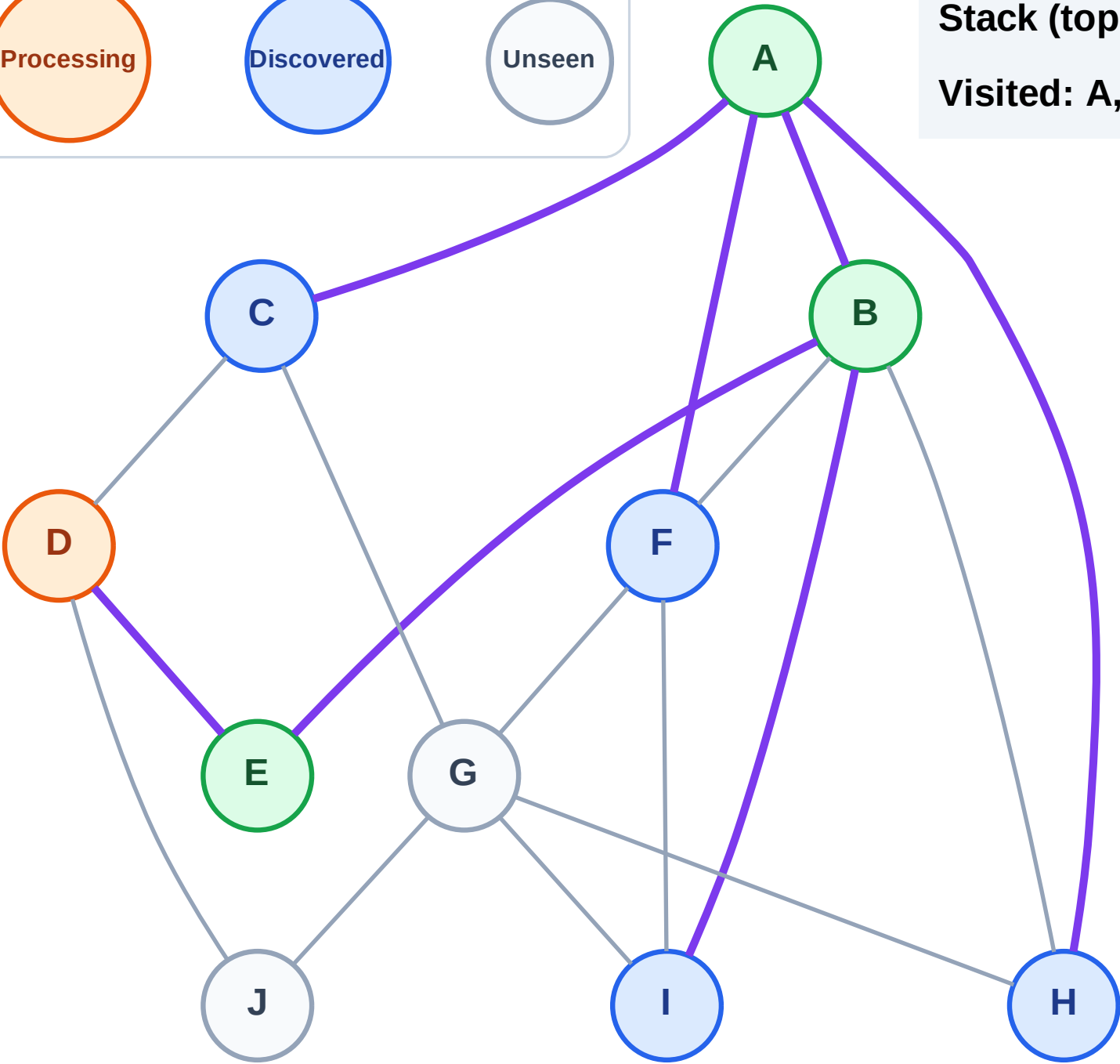
Processing

Discovered

Unseen

Stack (top at right): H | F | C | I

Visited: A, B, E



DFS Traversal

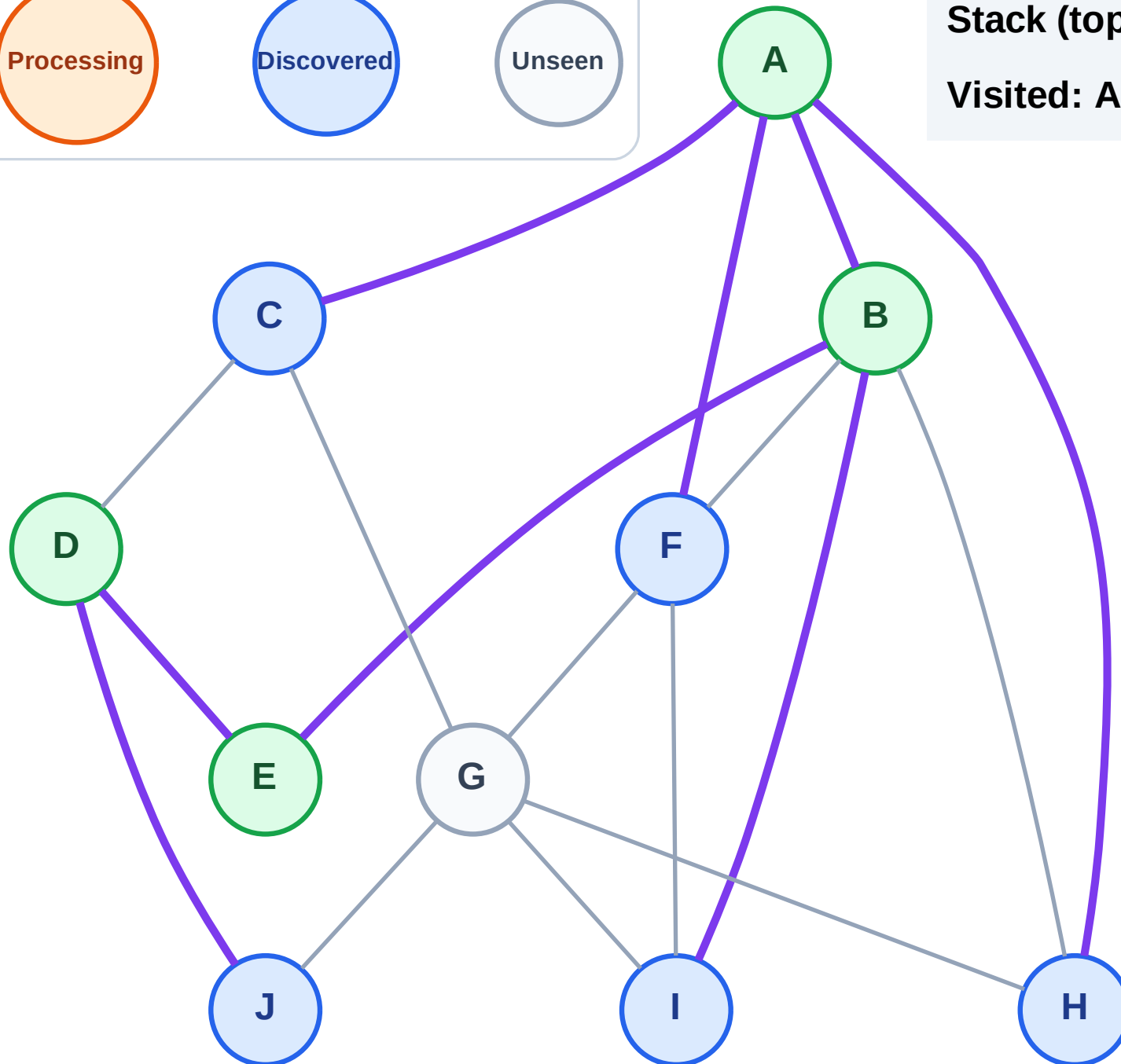
Step 9: Discover J from D

Legend



Stack (top at right): H | F | C | I | J

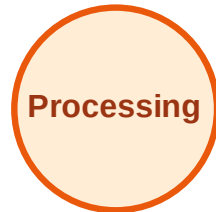
Visited: A, B, D, E



DFS Traversal

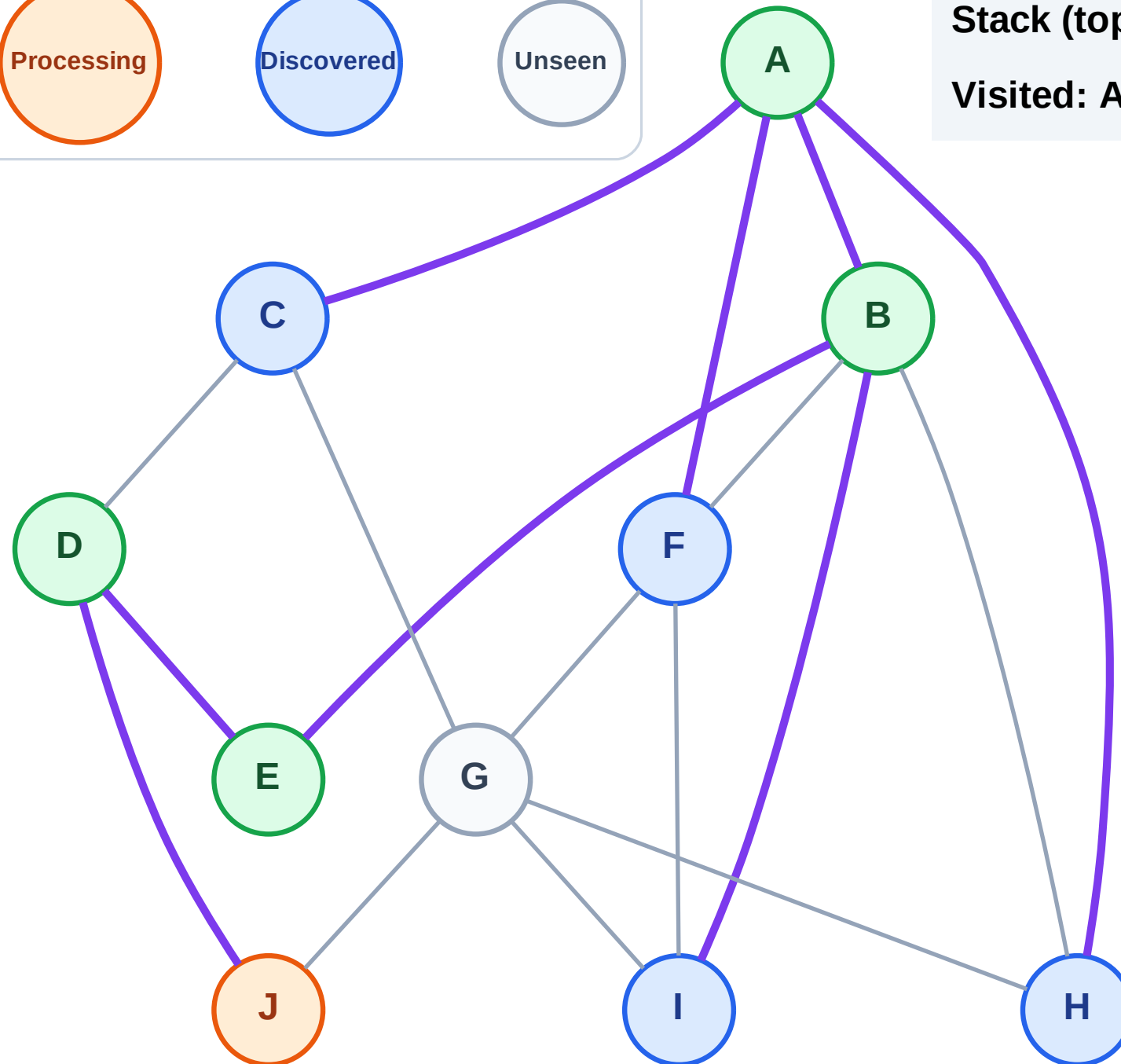
Step 10: Process node J

Legend



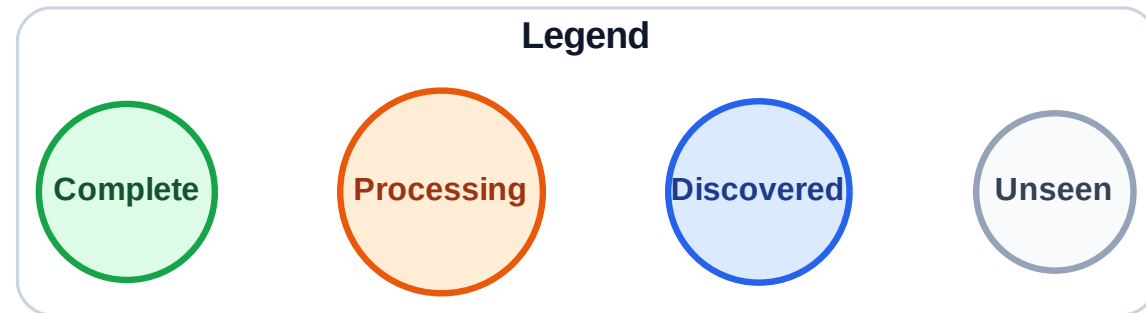
Stack (top at right): H | F | C | I

Visited: A, B, D, E



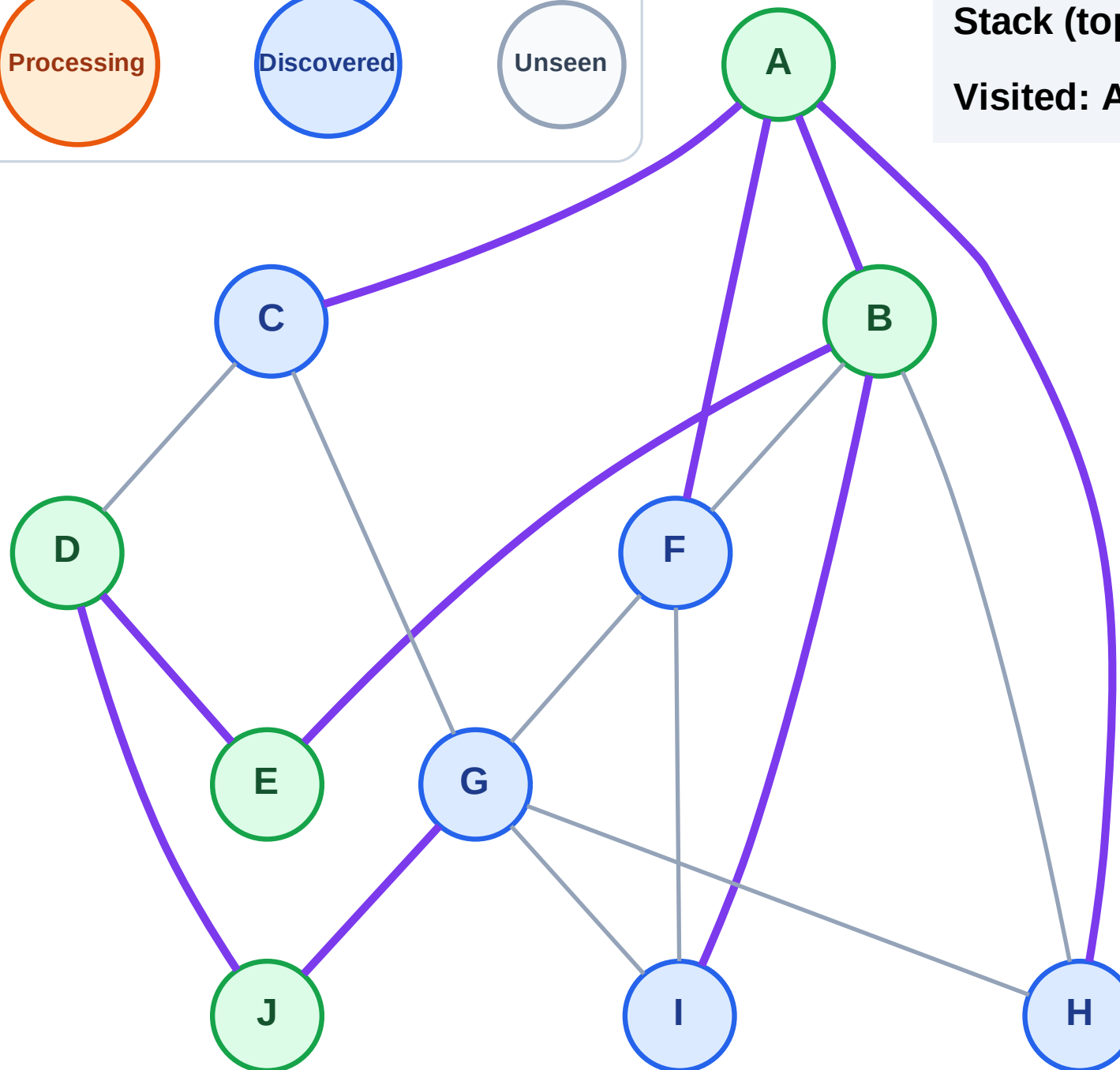
Step 11: Discover G from J

Step 11: Discover G from J



Stack (top at right): H | F | C | I | G

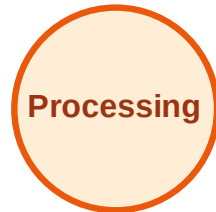
Visited: A, B, D, E, J



DFS Traversal

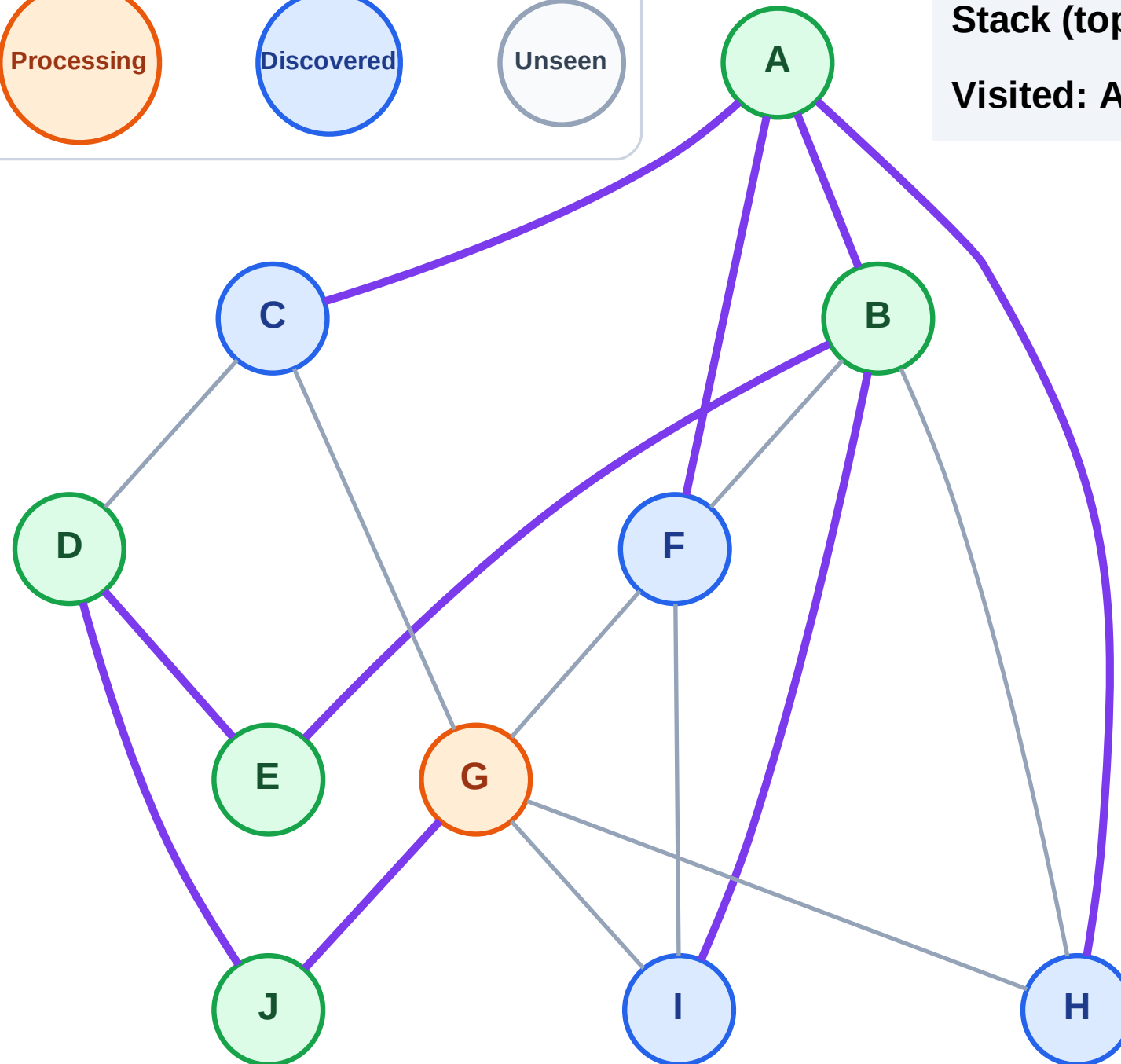
Step 12: Process node G

Legend



Stack (top at right): H | F | C | I

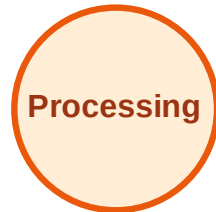
Visited: A, B, D, E, J



DFS Traversal

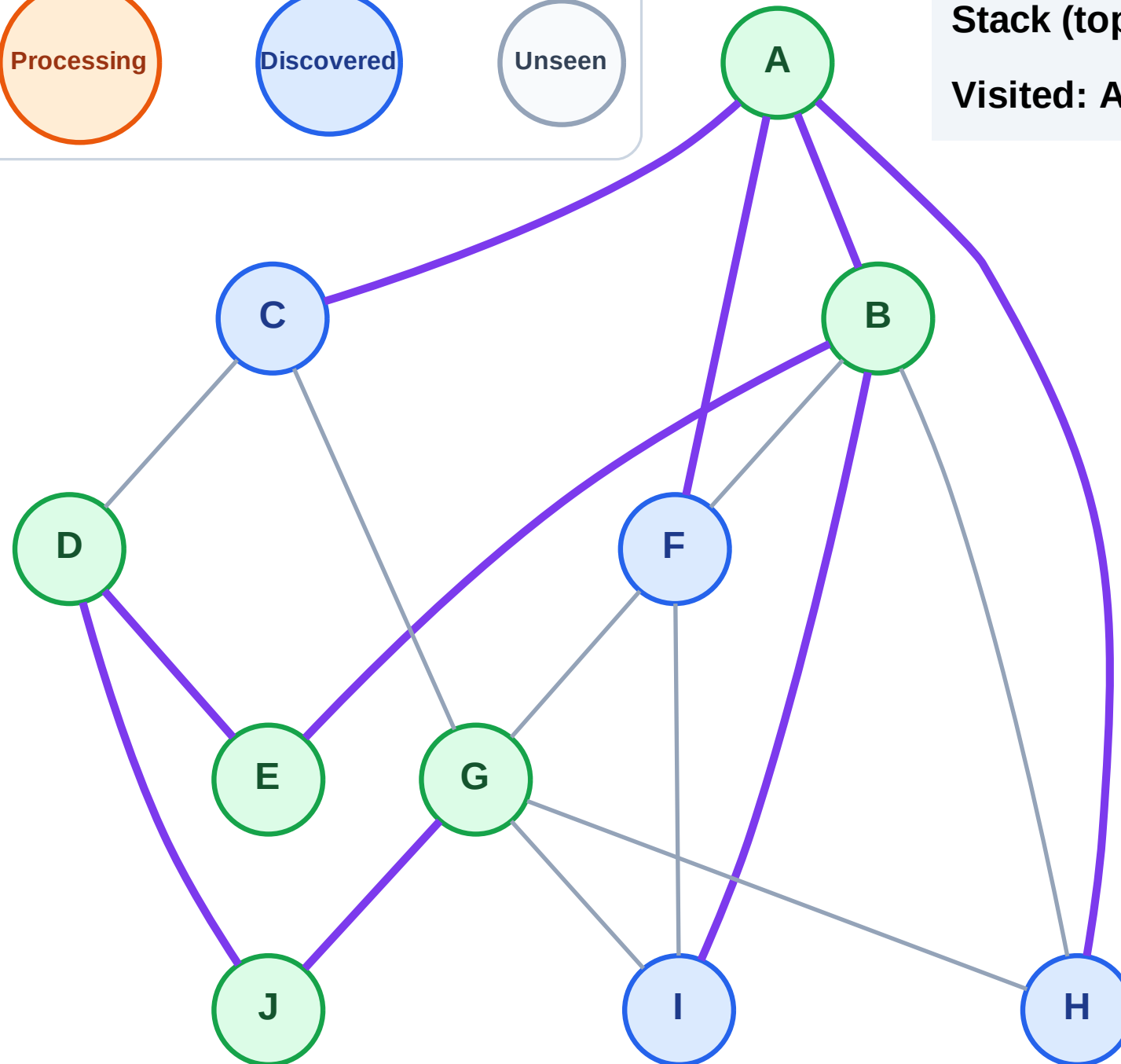
Step 13: No new neighbors from G

Legend



Stack (top at right): H | F | C | I

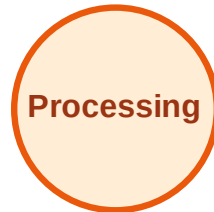
Visited: A, B, D, E, G, J



DFS Traversal

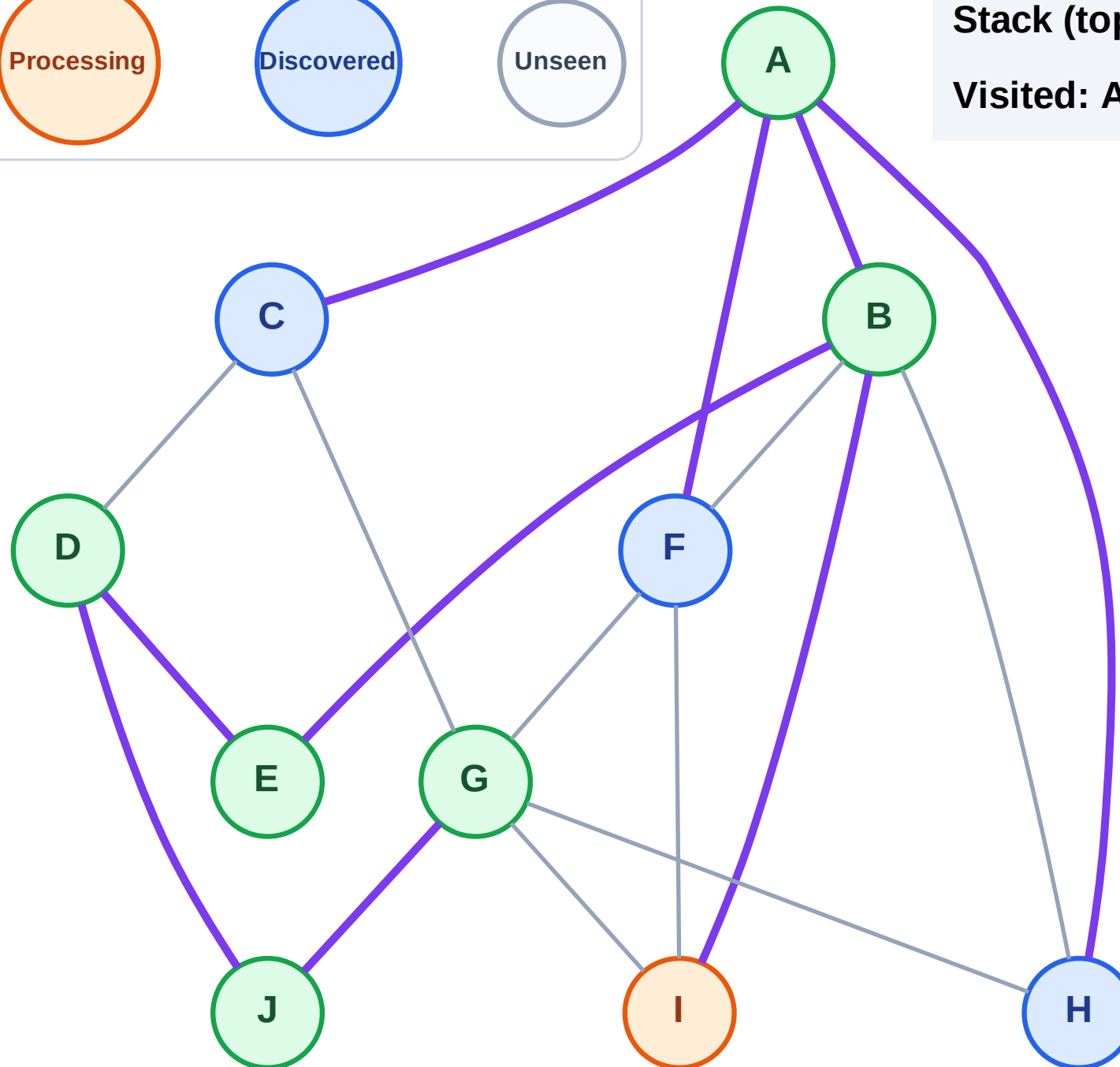
Step 14: Process node I

Legend



Stack (top at right): H | F | C

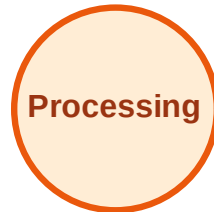
Visited: A, B, D, E, G, J



DFS Traversal

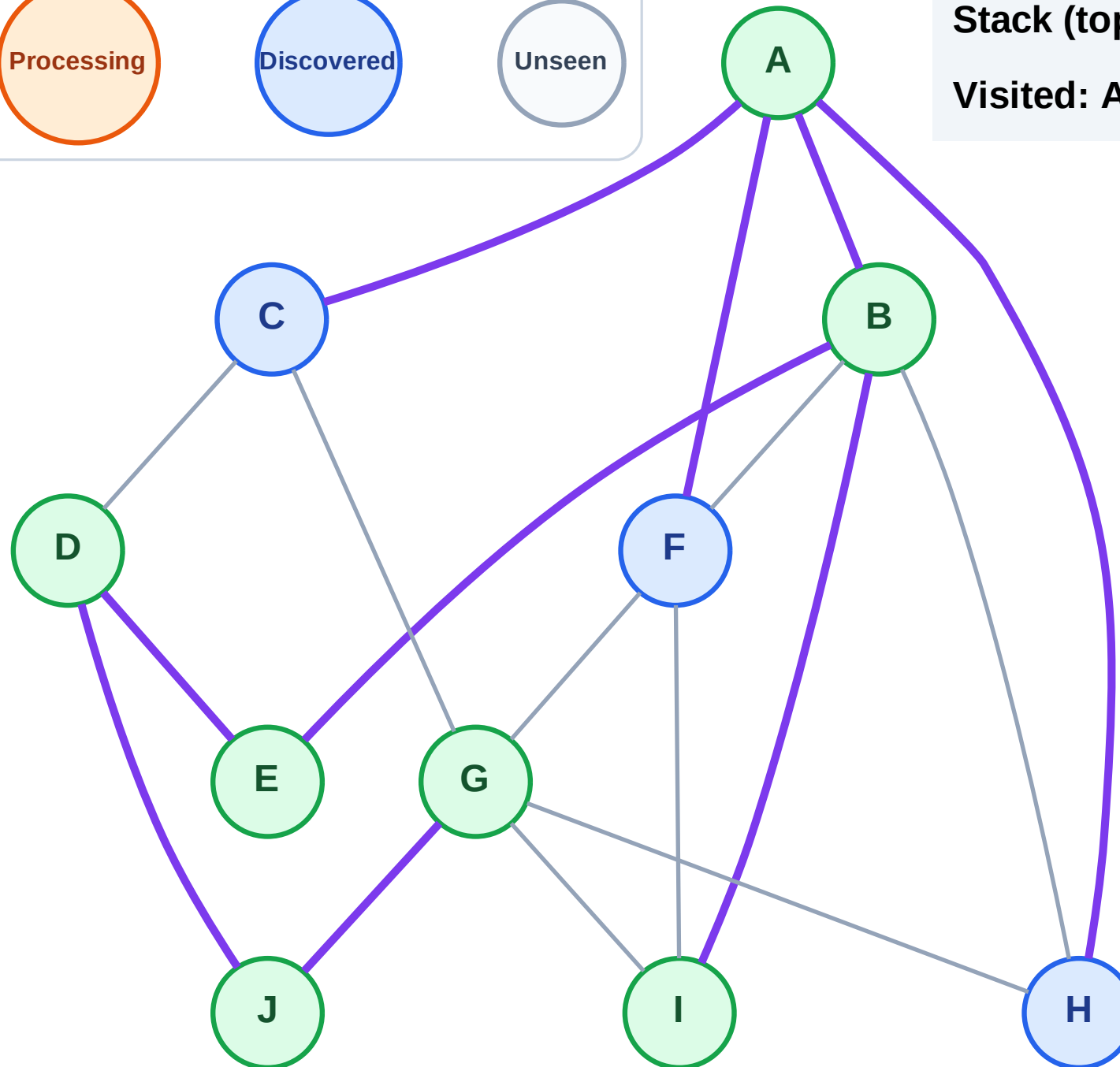
Step 15: No new neighbors from I

Legend



Stack (top at right): H | F | C

Visited: A, B, D, E, G, I, J



DFS Traversal

Step 16: Process node C

Legend

Complete

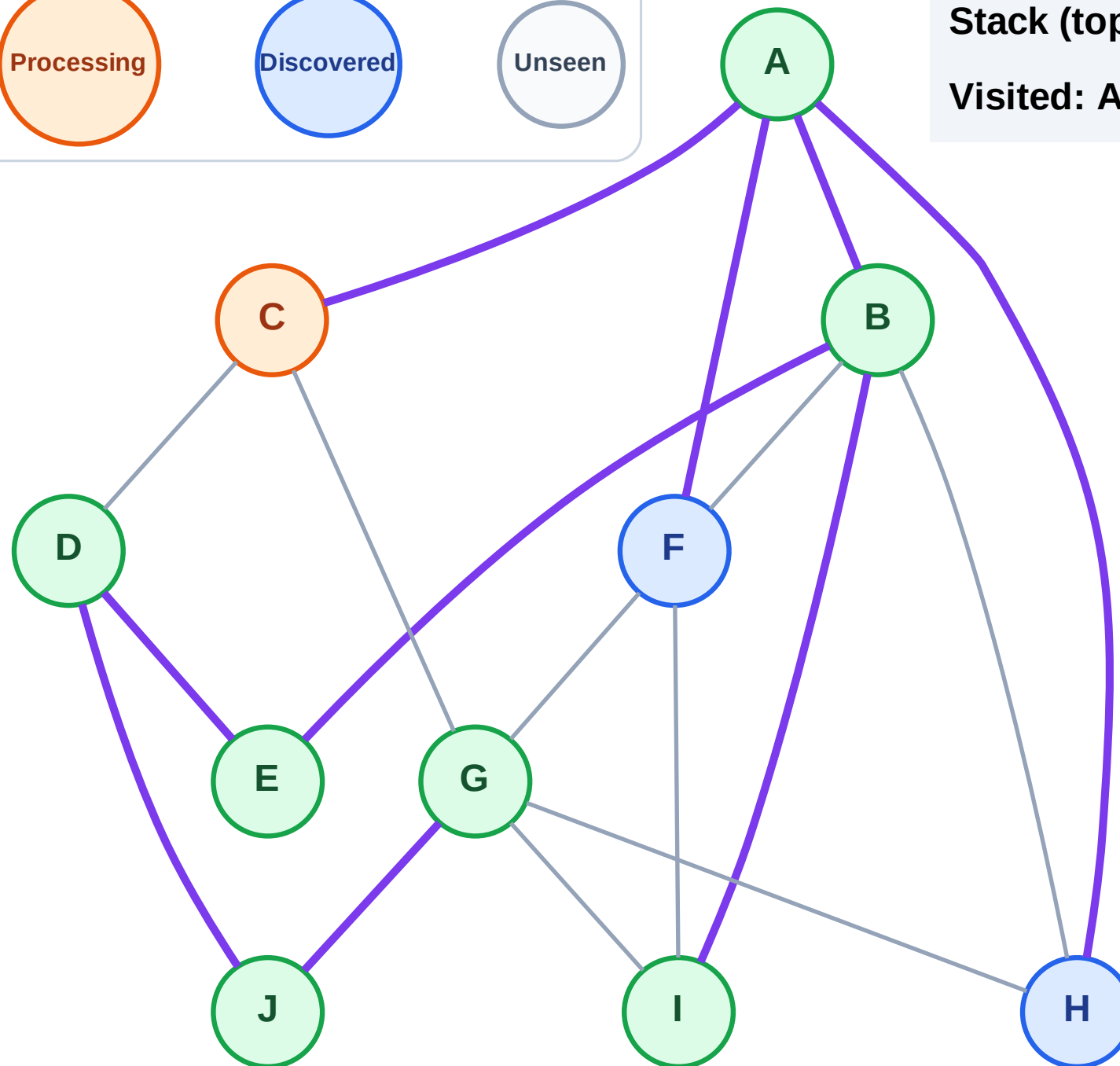
Processing

Discovered

Unseen

Stack (top at right): H | F

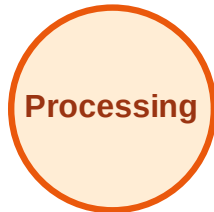
Visited: A, B, D, E, G, I, J



DFS Traversal

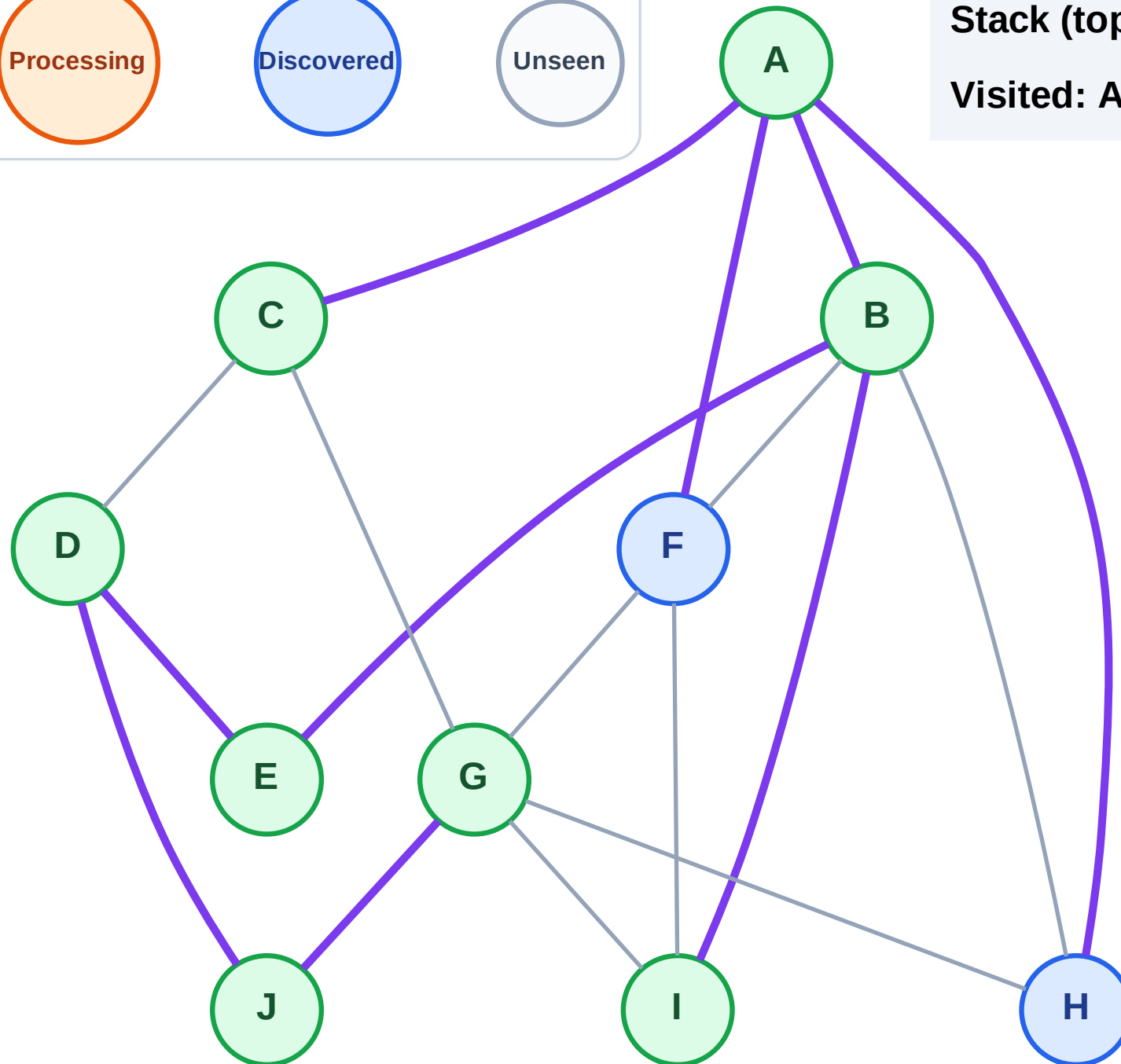
Step 17: No new neighbors from C

Legend



Stack (top at right): H | F

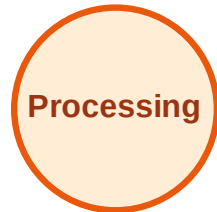
Visited: A, B, C, D, E, G, I, J



DFS Traversal

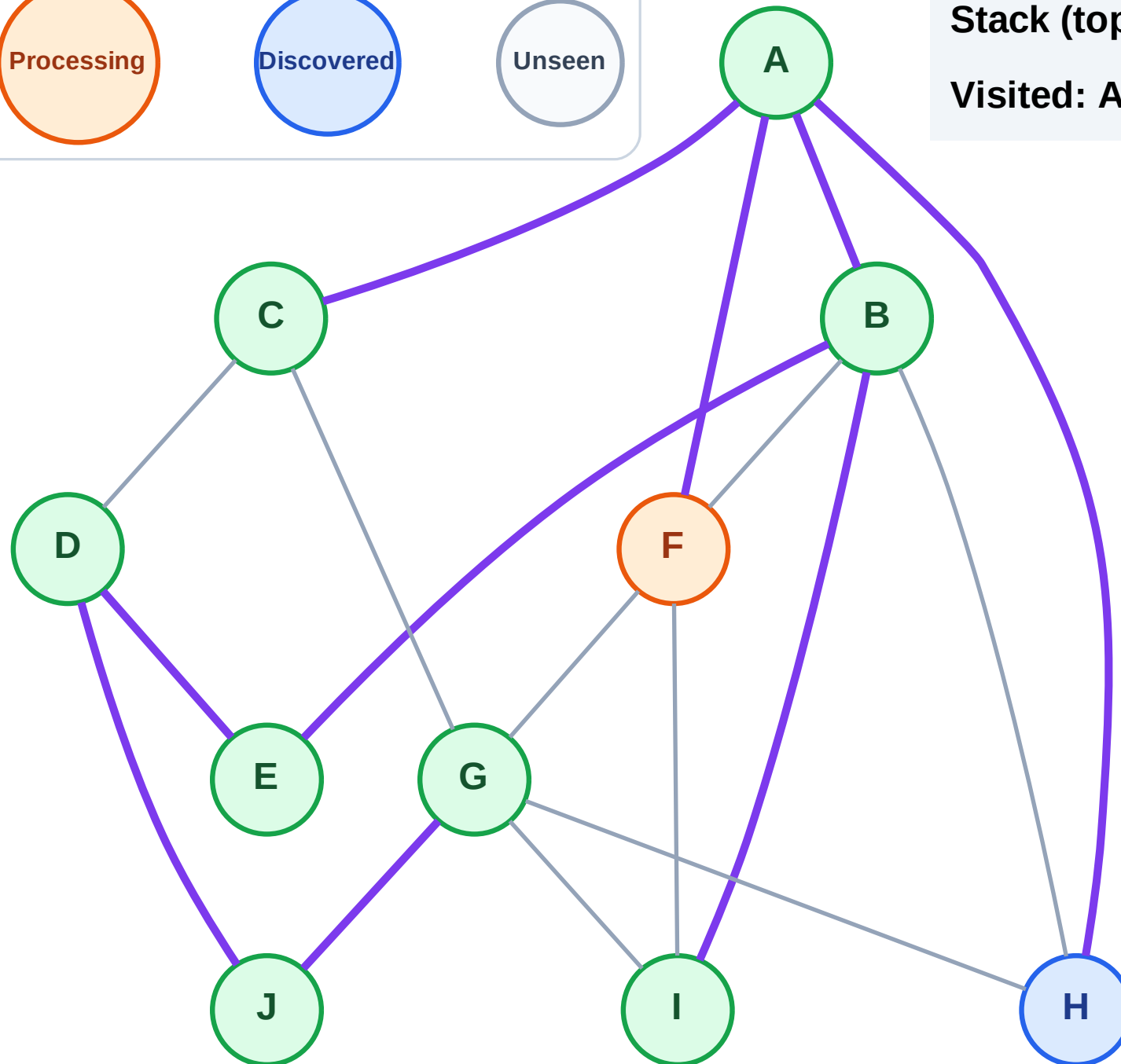
Step 18: Process node F

Legend



Stack (top at right): H

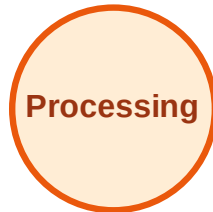
Visited: A, B, C, D, E, G, I, J



DFS Traversal

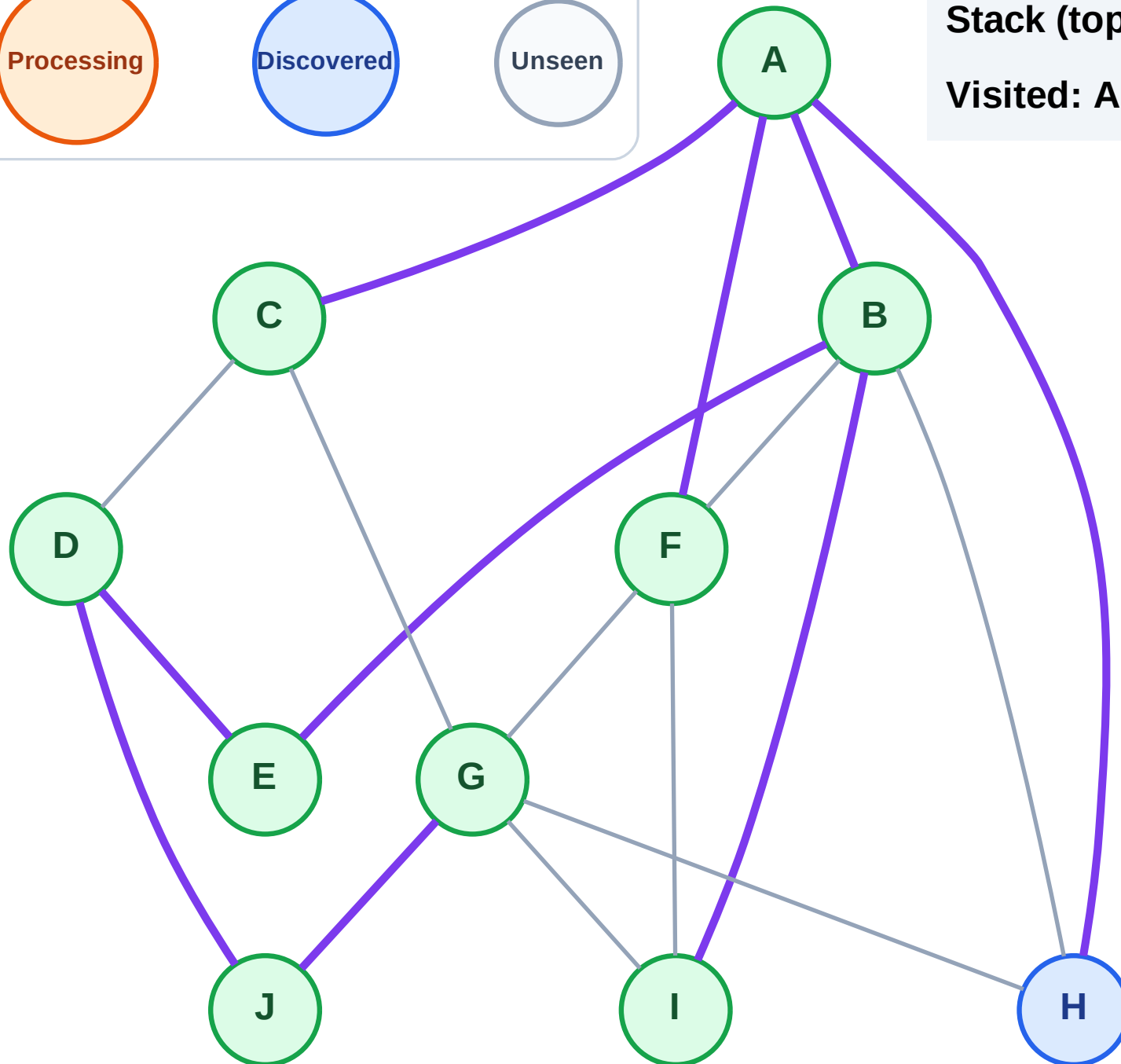
Step 19: No new neighbors from F

Legend



Stack (top at right): H

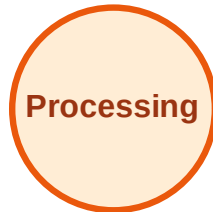
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

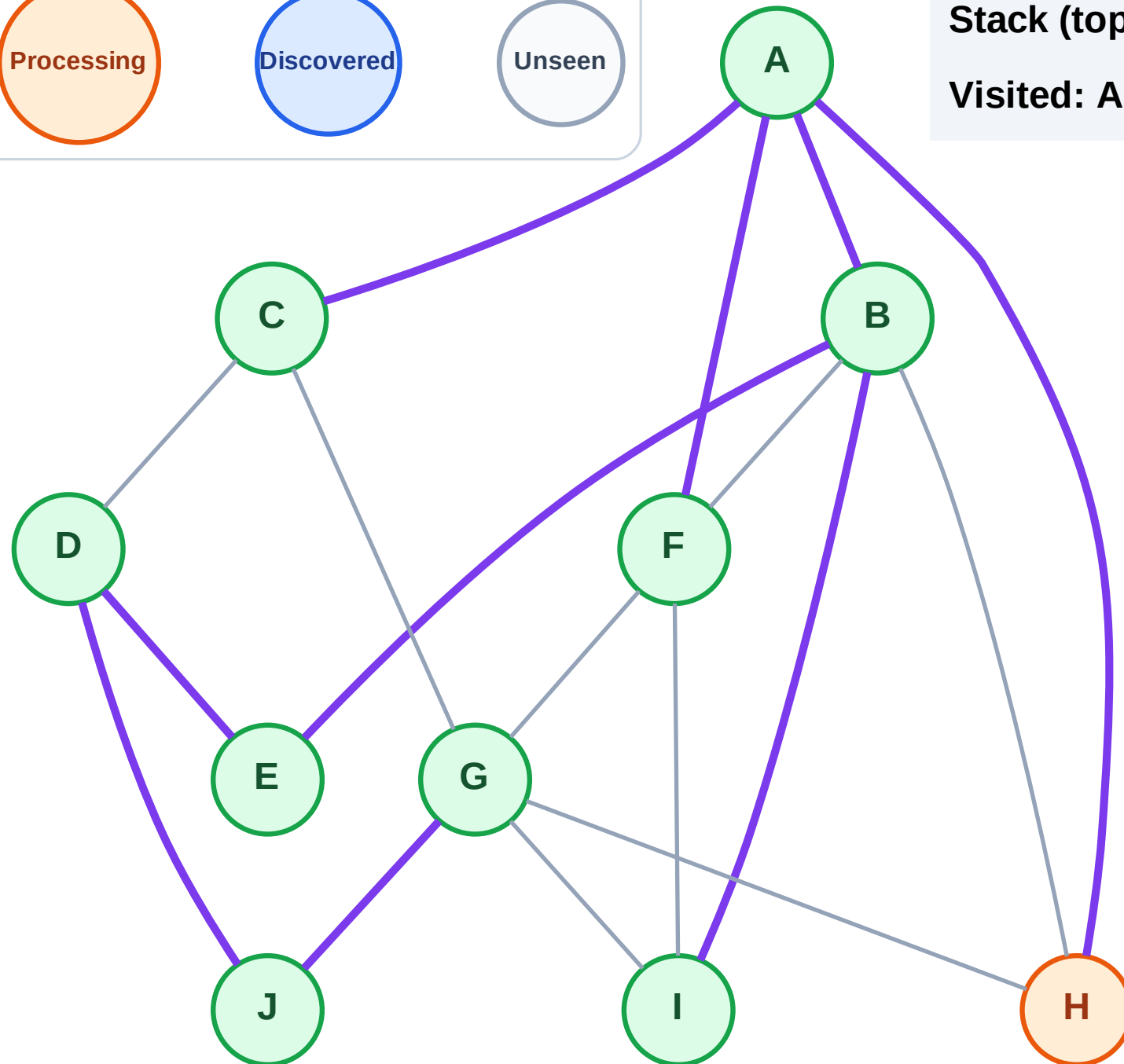
Step 20: Process node H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

